

Chapter 16 :

Oscillators

16.1 Introduction

So far we have studied amplifiers and feedback in amplifiers. In this chapter, we will study oscillators. Generally an oscillator is made by using an amplifier and a positive feedback network. Although amplifiers use negative feedback to provide stability, oscillators use positive feedback to generate undamped continuous oscillation. We know that an amplifier produces output if an input signal is applied. The output of the amplifier has same frequency and wave shape as the input signal. In fact an amplifier is an energy converter. It takes energy from DC power supply and increase the energy of the signal without changing its shape and frequency. When there is no input signal, there is no output signal also, that is, no energy conversion. On the other hand, an oscillator does not require any external signal as input. Without any input signal it can produce output signal so long as the DC power supply is connected. Hence, we can say: ***an oscillator is a circuit that generates an AC output signal without any external input signal or an oscillator is a circuit that converts DC energy into AC energy at very high frequency.*** The output of the oscillators may be sinusoidal or non-sinusoidal. Among the non-sinusoidal the most common waveforms are: square wave, triangular wave, saw-tooth wave, etc.

16.2 Classifications of Oscillators

Oscillators are classified into several types based on various factors- like shape of waveform, range of frequency, the parameters used, etc. The followings are the broad classification of oscillators.

16.2.1 According to the Generated Waveform

Based on the output waveform, oscillators are classified as,

Sinusoidal Oscillators: This type of oscillator generates sinusoidal current or voltages. They are most commonly used in electronics. Examples of some sinusoidal oscillators are

- Tuned-circuits or LC feedback oscillators such as Hartley, Colpitts and Clapp etc.
- RC phase-shift, Wien-bridge oscillators etc.
- Negative-resistance oscillators such as tunnel diode oscillator.
- Crystal oscillators such as Pierce oscillator.
- Heterodyne or beat-frequency oscillator (BFO).

Non-sinusoidal Oscillators: This type of oscillators generates non-sinusoidal output. Among various non-sinusoidal oscillators, the most common waveforms are triangular wave, square wave, rectangular wave, saw tooth wave etc.

16.2.2 According to the Frequency Selecting Network

Every oscillator has a circuit that selects the frequency of oscillation. Depending on the components of the frequency selecting network, oscillators are classified as:

LC Oscillators: The oscillator using inductor (L) and capacitor (C) components as the frequency selecting network is called as LC oscillator. These oscillators can generate frequencies in the range of. LC oscillators are not used to generate very low frequency, because to generate very low frequency the size of inductors becomes impractically large.

RC Oscillators: The oscillator using resistance and capacitor components is called as RC oscillators. These oscillators can generate frequencies in the range of .

Crystal Oscillators: Crystals are used as frequency selecting elements in some oscillators. Crystal oscillators can provide a very stable frequency. The frequency range of crystal oscillators is.

Negative Resistance Oscillators: In some oscillators, instead of using positive feedback network, the negative resistance properties of some devices are used. There are some devices, like UJT (Uni-junction Transistor), Gunn diode, SCR (Silicon Control Rectifier) etc., show negative resistance in their characteristics. The negative resistance helps to compensate the loss of an oscillator and thus sustains continuous oscillation. These oscillators can generate frequencies in the range of .

16.2.3 According to the Frequency Generated

Oscillators can be used to produce the waveforms at frequencies ranging from low to very high values. Depending on the frequency of the output waveform, oscillators can be classified as:

Low Frequency Oscillators: This type of oscillators generates very low frequency. The frequency range of this oscillator is below approximately 20 Hz. Low-frequency oscillators (LFO) are used to create rhythmic pulses or sweeps. These pulses or sweeps are used to modulate musical equipment like synthesizers to create audio effects such as vibrato, tremolo and phasing.

Audio Frequency Oscillators: An audio oscillator produces frequencies in the audio range, that is, about 16 Hz to 20 kHz. RC oscillators are used to generate audio frequency range.

RF Frequency Oscillators: An RF oscillator produces signals in the radio frequency (RF) range of about 100 kHz to 100 GHz. Generally, LC and crystal oscillators are used to generate frequencies in this range.

Microwave Oscillators: Microwave oscillators are also RF oscillator. However, frequency range 800 MHz to 300 GHz is referred to as microwave. So, the oscillator that generates frequencies in this range is called microwave oscillators. Special types of electron tubes or semiconductor devices are used to generate microwave signals. Negative-resistance oscillators are also used to generate frequencies in microwave range and above, since at these frequencies feedback oscillators perform

16.2.4 Based on the Usage of Feedback

The oscillators consisting of feedback network to satisfy the required conditions of the oscillations are called as feedback oscillators. Whereas, the oscillators that have no feedback networks are called non-feedback oscillators. The UJT relaxation oscillator is the example of non-feedback oscillator which uses a negative resistance region of the characteristics of the device.

16.3 Oscillators Theory

The main statement of the oscillator is that the oscillation is achieved through positive feedback which generates the output signal without input signal. Also, the voltage gain of the amplifier increases with the increase in the amount of positive feedback.

In order to understand this concept, let us consider a non-inverting amplifier with a voltage gain A_v and a positive feedback network with feedback factor of β as shown in Fig.16-1.

Let us assume that a sinusoidal input signal v_s is applied at the input. Since the amplifier is non-inverting, the output signal v_o is in phase with v_s . A feedback network feeds a part of v_o to the input and the amount fed back (v_f) depends on the feedback network gain (also called feedback factor) β .

In Chapter 12, we have found that the gain of an amplifier with positive feedback is,

$$A_f = \frac{A_v}{1 - A_v\beta} \quad (16-1)$$

Where, $A_v\beta$ (called *loop-gain*) is the product of open-loop gain and the feedback factor. If the value of loop-gain is unity, that is if $A_v\beta = 1$, from Equ.(16-1) we find that the gain of the amplifier with feedback will be infinity, that is $A_f = A_v/(1 - 1) = \infty$. So, if the signal source is disconnected ($v_s = 0$), the amplifier will produce a finite output. The circuit can generate output just by feeding a part of the output as its own input. In this condition, the amplifier will work as an oscillator and will generate output signal without any input.

When an oscillator is powered ON some sort of noise, like transient noise, is produced. Though noise has a wide frequency range, only a particular frequency, for which the oscillator is designed, will fulfill the condition of oscillation. At first the gain of the amplifier will be a little larger than $1/\beta$ (or $A_v\beta > 1$). So, that particular frequency will gradually be amplified as shown in Fig.16-2. Due to the saturation of the amplifier the output will be fixed to a certain value, and the gain of the amplifier will be adjusted to $A_v = 1/\beta$. Sometimes automatic gain control (AGC) circuit is used to limit the amplitude of the oscillators.

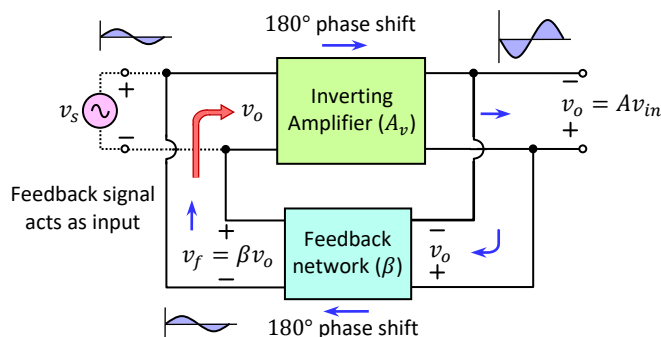


Fig.16-1: Amplifier with positive feedback

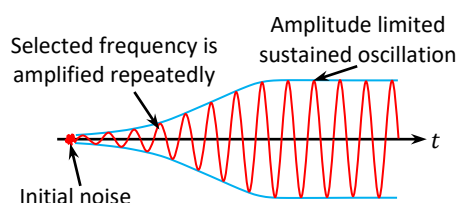


Fig.16-2: Development of output signal in an oscillator

16.3.1 Barkhausen Criteria

We have already discussed that an amplifier can work as an oscillator if we add positive feedback to it. Actually the amplifier will work as an oscillator under some conditions. Conditions which are

required to be satisfied to operate an amplifier as an oscillator are called as *Barkhausen criteria* for sustained oscillations. It was put forth in 1921 by German physicist Heinrich Georg Barkhausen (1881–1956)^[1].

The Barkhausen criteria should be satisfied by an amplifier with positive feedback to ensure the sustained oscillations. In fact, Barkhausen criteria are nothing but formal representation of the above theory of oscillator.

The Barkhausen criteria state the following:

1. The value of the loop-gain of the amplifier must be equal to unity, that is, $|A_v\beta| = 1$. In fact, at first (during the buildup of output signal) $A_v\beta$ should be slightly greater than unity and when the output signal is established, $A_v\beta$ should be unity. So we can write $A_v\beta \geq 1$.
2. The phase shift around the loop should be zero or an integer multiple of 360° , that is $\angle A_v\beta = n \times 360^\circ$, where $n = 0, 1, 2$ etc. This criterion ensures the positive feedback.

16.4 RC OSCILLATOR

The RC phase shift oscillator is used to generate a wide range of frequencies, from few Hz to 200 kHz. Fig.16-3 shows the circuit diagram of a RC phase shift oscillator. The common emitter amplifier is biased using a voltage divider network (R_1, R_2). R_E is used to stabilize the DC operating point and R_C is the load resistance of the amplifier. C_{oc} is the output coupling capacitor and C_E is the emitter bypass capacitor. This CE amplifier provides necessary gain and 180° phase shift. To obtain a positive feedback, essential for oscillations, the frequency determining circuit must introduce another 180° phase shift. This phase shift of 180° is obtained by cascading three RC sections. Here, the RC sections also work as frequency selecting network. Phase shift occurs, because in RC circuit due to the capacitor the current leads the applied voltage by an angle $\phi = \tan^{-1}(X_C/R)$. So the voltage across R will lead the input voltage. The smaller is the capacitance the more will the current lead the voltage for a given resistance. Theoretically a RC circuit can produce maximum 180° phase shift. So, we can produce 360° phase shift using two RC sections. But in that case, the value of X_C should be very large (infinity), so the voltage across the resistor will be very low (zero). For this reason, as shown in Fig.16-4, three RC sections are used and each section is designed to produce 60° phase shift. With a proper choice of R and C, total 180° phase shift is achieved for the desired frequency. At the third stage, we have used $R_3 - h_{ie}$ because the input impedance of the amplifier $Z_{in} \approx h_{ie}$ will be added with this resistor.

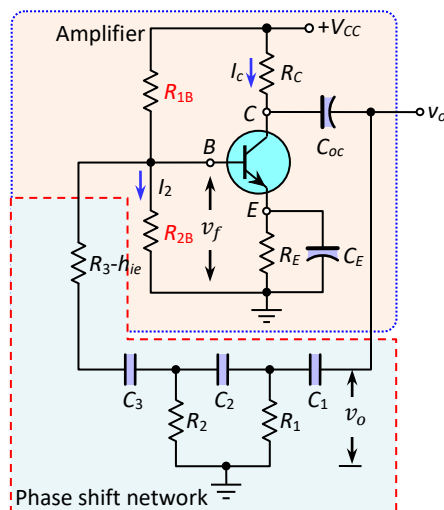


Fig.16-3: RC phase shift oscillator

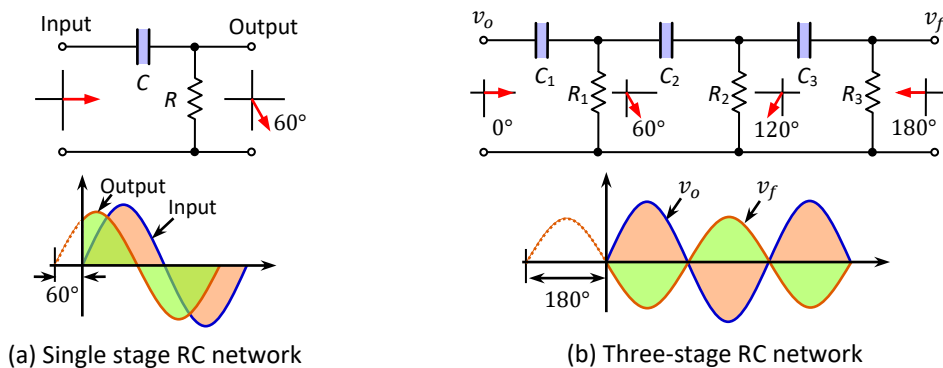


Fig.16-4: Phase shift mechanism in RC networks

16.4.1 Circuit Operation:

At first the gain of the amplifier (A_v) will be greater than the feedback factor (β), that is $A_v\beta > 1$. When the power is applied to the circuit, electrical noise in the circuit or the turn-ON transient provides an initial signal to start oscillation. The noise has many frequencies. But only a particular frequency, for which the amplifier has been designed, will get 180° phase shifted by the RC network and will be applied to amplifier's input. So, that frequency will be amplified by the amplifier. Now RC network will get an amplified signal, so it will feed a larger signal at the input. Due to this larger input signal, the amplifier will produce more amplified signal at the output. As $A_v > 1/\beta$, a positive regenerative process will continue. When the generated signal attains a substantial amplitude, due to the little nonlinearity of BJT, A_v will decrease. Thus a signal of desired frequency with constant amplitude is produced in the circuit.

16.4.2 Analysis:

To analyze the oscillator circuit we will assume three RC sections identical: $C_1 = C_2 = C_3 = C$, and $R_1 = R_2 = h_{ie} = R$, that is, no additional resistor in place of R_3 . The input impedance ($Z_{in} \approx h_{ie}$) of the amplifier will work as R_3 . If we use $R_3 \neq 0$, the feedback voltage will be low because it will be divided between R_3 and h_{ie} . We will consider the output section of the amplifier as a dependent voltage source v_o and a series resistance Z_o . For a CE amplifier $Z_o \approx R_C$. With all these considerations, we can represent the feedback network as shown in Fig.16-5.

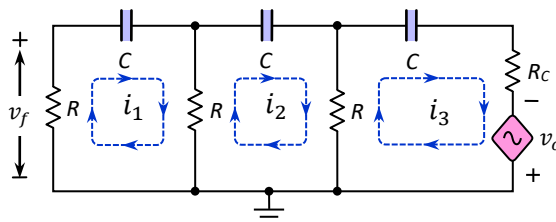


Fig.16-5: RC phase shift network with three loop currents

Now applying KVL in three loops of Fig.16-5, we get,

$$\begin{aligned} (2R + 1/j\omega C)i_1 - Ri_2 + 0i_3 &= 0 \\ -Ri_1 + (2R + 1/j\omega C)i_2 - Ri_3 &= 0 \\ 0i_1 - Ri_2 + (R + R_C + 1/j\omega C)i_3 &= v_o \end{aligned}$$

Considering $j\omega = s$ and writing the above equations in matrix form we get,

$$\begin{bmatrix} (2R + 1/sC) & -R & 0 \\ -R & (2R + 1/sC) & -R \\ 0 & -R & (R + R_C + 1/sC) \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ v_o \end{bmatrix}$$

To calculate i_1 , we will use the Crammer's rule. The value of Δ will be,

$$\Delta = \begin{vmatrix} (2R + 1/sC) & -R & 0 \\ -R & (2R + 1/sC) & -R \\ 0 & -R & (R + R_C + 1/sC) \end{vmatrix}$$

$$\text{or,} \quad \Delta = (4R^2 + 4R/sC + 1/s^2C^2)(R + R_C + 1/sC) - R^2(2R + 1/sC) - R^2(R + R_C + 1/sC)$$

$$\text{or,} \quad \Delta = (3R^2 + 4R/sC + 1/s^2C^2)(R + R_C + 1/sC) - R^2(2R + 1/sC)$$

$$\text{or,} \quad \Delta = (3R^3 + 4R^2/sC + R/s^2C^2) + (3R_CR^2 + 4RR_C/sC + R_C/s^2C^2) + (3R^2/sC + 4R/s^2C^2 + 1/s^3C^3) - 2R^3 - R^2/sC$$

$$\text{or,} \quad \Delta = (R^3 + 6R^2/sC + 5R/s^2C^2 + 1/s^3C^3) + (3R_CR^2 + 4RR_C/sC + R_C/s^2C^2)$$

$$\Delta_1 = \begin{vmatrix} 0 & -R & 0 \\ 0 & (2R + 1/sC) & -R \\ v_o & -R & (R + R_C + 1/sC) \end{vmatrix} = v_o R^2$$

$$\therefore i_1 = \frac{\Delta_1}{\Delta} = \frac{v_o R^2}{(R^3 + 6R^2/sC + 5R/s^2C^2 + 1/s^3C^3) + (3R_CR^2 + 4RR_C/sC + R_C/s^2C^2)}$$

The feedback voltage will be,

$$v_f = i_1 R = \frac{v_o R^3}{(R^3 + 6R^2/sC + 5R/s^2C^2 + 1/s^3C^3) + (3R_CR^2 + 4RR_C/sC + R_C/s^2C^2)}$$

$$\text{or,} \quad \beta = \frac{v_f}{v_o} = \frac{1}{(1 + 6/sRC + 5/s^2R^2C^2 + 1/s^3R^3C^3) + (3R_C/R + 4R_C/sR^2C + R_C/s^2R^3C^2)}$$

As we have assumed $s = j\omega$, $\therefore 1/s = -j/\omega$, $1/s^2 = -1/\omega^2$, and $1/s^3 = j/\omega^3$. Putting these values in the above equation,

$$\beta = \frac{1}{(1 - 6j/\omega RC - 5/\omega^2 R^2 C^2 + j/\omega^3 R^3 C^3) + (3R_C/R - 4jR_C/\omega R^2 C - R_C/\omega^2 R^3 C^2)}$$

$$\therefore \beta = \frac{1}{(1 - 5/\omega^2 R^2 C^2 + 3R_C/R - R_C/\omega^2 R^3 C^2) - j/\omega RC(6 - 1/\omega^2 R^2 C^2 + 4R_C/R)} \quad (16-2)$$

The feedback factor must have 180° phase shift. So the imaginary part in Equ.(16-2) must be zero at the frequency of oscillation ($\omega = \omega_o$).

$$\therefore -j/\omega_o RC \left(6 - \frac{1}{\omega_o^2 R^2 C^2} + \frac{4R_C}{R} \right) = 0$$

$$\text{or,} \quad 6 + 4R_C/R = \frac{1}{\omega_o^2 R^2 C^2}$$

$$\text{or,} \quad \omega_o^2 = \frac{1}{R^2 C^2 (6 + 4R_C/R)} \quad (16-3)$$

$$\therefore \quad \boxed{f_o = \frac{1}{2\pi RC \sqrt{(6 + 4R_C/R)}}} \quad (16-4)$$

Equ.(16-4) is the expression for frequency of oscillation of phase shift oscillator. If $R_C \ll R$,

$$\therefore \quad \boxed{f_o = \frac{1}{2\pi RC \sqrt{6}}} \quad (16-5)$$

Now considering the imaginary term zero in Equ.(16-2), and putting the value of ω_o^2 from Equ.(16-3) we get,

$$\beta = \frac{1}{1 - 5(6 + 4R_C/R) + 3R_C/R - R_C(6 + 4R_C/R)/R}$$

$$\text{or,} \quad \beta = \frac{1}{1 - 30 - 20R_C/R + 3R_C/R - 6R_C/R - 4R_C^2/R^2}$$

$$\therefore \quad \boxed{\beta = -\frac{1}{29 + 23R_C/R + 4R_C^2/R^2}} \quad (16-6)$$

Equ.(16-6) is the expression for feedback factor of phase shift oscillator.

To satisfy the condition $A_v \beta \geq 1$, the voltage gain of the amplifier should be,

$$|A_v| \geq |1/\beta|$$

$$\therefore \quad \boxed{|A_v| \geq 29 + \frac{23R_C}{R} + \frac{4R_C^2}{R^2}} \quad (16-7)$$

If R_C is negligibly small compared to R , then we get,

$$\therefore \quad \boxed{|A_v| \geq 29} \quad (16-8)$$

But, if we select $R_C = R$, then

$$\boxed{|A_v| \geq 56} \quad (16-9)$$

Example 16-1

Design a phase shift oscillator to generate 25 kHz signal. Assume that the supply voltage is 12 V, and $h_{fe} = 150$.

Solution:

Let $I_C = 1 \text{ mA} \approx I_E$, and $V_E = 3 \text{ V}$.

$$\therefore \quad R_E = \frac{V_E}{I_E} = \frac{3 \text{ V}}{1 \text{ mA}} = 3 \text{ k}$$

Let the current through R_4 be $I_4 = I_C/10 = 0.1 \text{ mA}$. [See Fig.16-6]

$$\therefore R_4 = \frac{V_E + V_{BE}}{I_4} = \frac{3 \text{ V} + 0.7 \text{ V}}{0.1 \text{ mA}} = 37 \text{ k}, \text{ use standard } 36 \text{ k}$$

$$\therefore R_3 = \frac{V_{CC} - V_{R2}}{I_4 + I_C/h_{fe}} = \frac{12 \text{ V} - 3.7 \text{ V}}{0.1 \text{ mA} + 1 \text{ mA}/150} = 77.8 \text{ k}, \text{ use standard } 75 \text{ k}$$

$$\text{Using Equ.(16-111)} \quad r_e = \frac{26 \text{ mV}}{I_E} \approx \frac{26 \text{ mV}}{1 \text{ mA}} = 26 \Omega$$

$$\therefore h_{ie} = h_{fe} \times r_e = 150 \times 26 \Omega = 3.9 \text{ k}$$

We will consider $R_1 = R_2 = h_{ie} = R = 3.9 \text{ k}$. Using Equ.(16-5),

$$C = \frac{1}{2\pi R f_o \sqrt{6}} = \frac{1}{2\pi \times 3.9 \text{ k} \times 25 \text{ k} \times \sqrt{6}} \\ = 666 \text{ pF } 833 \text{ pF, use standard } 820 \text{ pF}$$

Voltage gain of the CE amplifier is R_C/r_e . Putting this value in Equ.(16-7),

$$\frac{R_C}{0.026} = 29 + \frac{23R_C}{3.9} + \frac{4R_C^2}{3.9^2}$$

Solving this equation, we will find $R_C = 0.9 \text{ k}$, but actual value of R_C should be little bit larger. I have simulated the circuit using Ltpice. For $R_C = 1.4 \text{ k}$ the circuit starts to oscillate but amplitude is small. For $R_C = 3 \text{ k}$, $\pm 2.8 \text{ V}$ signal is found.

We can calculate the values of C_{oc} and C_E . But a thumb rule is $C_E = C_{oc} > 10C$. Let us use 10 nF capacitors for C_E and C_{oc} .

Results: The designed circuit has been simulated using Ltpice and is given in Fig.15.6. The output signal is also given in this figure.

Comments: According to Equ.(16-6) feedback factor also decreases with the increase of R_C . That increases the required gain of the amplifier.

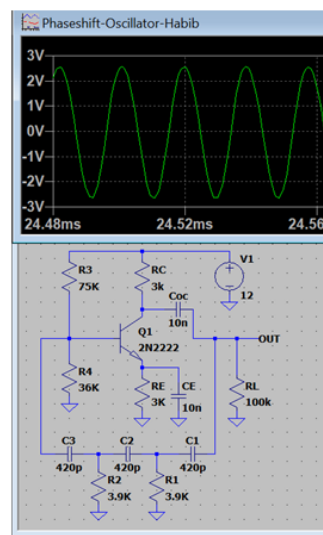


Fig.16-6: RC phase shift oscillator of Example 15-1 simulated in Ltpice

16.5 Phase Shift Oscillator Using Operational Amplifier

Phase shift oscillator can be made using operational amplifier. Fig.16-7 shows a phase shift oscillator using inverting operational amplifier. The inverting amplifier produces amplification and 180° phase shift and another 180° phase shift is produced by the 3-RC stages as described for BJT phase shift oscillator. R_3 is the input impedance of the amplifier as well as the resistance of the third RC stage. For simplicity of analysis, we will assume $R_1 = R_2 = R_3 = R$, and $C_1 = C_2 = C_3 = C$. Then, the frequency of oscillation can be calculated using Equ.(16-4),

Equ.(16-4),

$$f_o = \frac{1}{2\pi RC \sqrt{6 + 4R_C/R}}$$

Here, R_C will be the output impedance of operational amplifier. But we know that op-amp has very low output impedance, so we will neglect it. Thus considering $R_C = 0$, the above equation reduces to,

$$f_o = \frac{1}{2\pi RC \sqrt{6}}$$

Same as Equ.(16-5). By rearranging the above equation, we get $X_C = \sqrt{6}R$. This is a condition to get frequency of oscillation, that is to achieve 180° phase reversal in the phase shift network.

The value of feedback factor can be calculated using Equ.(16-6) as,

$$\beta = -\frac{1}{29 + 23R_C/R + 4R_C^2/R^2}$$

Again considering $R_C = 0$,

$$\beta = -\frac{1}{29} \quad (16-10)$$

Therefore, the gain of the inverting amplifier should be,

$$A_v = -\frac{R_f}{R_3} \geq \frac{1}{\beta} = -29$$

$\therefore$

$$R_f \geq 29R_3$$

16.5.1 Circuit Design

Design of phase shift oscillator with op-amp is comparatively easier than BJT phase shift oscillator. At first we have to select suitable value of the capacitance of the phase shift network that is the value of $C_1 = C_2 = C_3 = C$. The capacitance should be at least ten times greater than stray capacitance. Then using Equ.(16-5) the value of $R_1 = R_2 = R_3 = R$ has to be calculated for a required frequency of oscillation. Then the value of R_f should be set little greater than $29R_3$ to satisfy Barkhausen criteria. The operational amplifier must have sufficient slew rate to produce the desired frequency of oscillation. Moreover, the value of R should be very larger than the output impedance, and very lower than input impedance of op-amp. The value of C should be very larger than the parasitic capacitance of op-amp. A good choice is to use C not less than 1 nF. The process is further illustrated in Example 15-2.

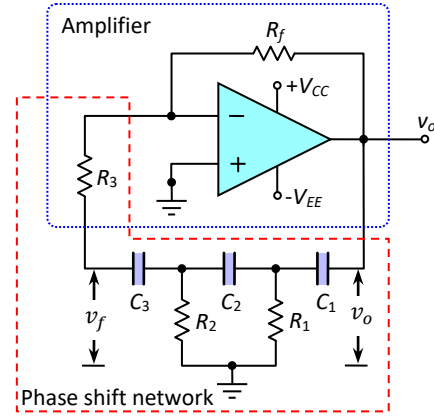


Fig.16-7: RC phase shift oscillator using op-amp

Example 16-2

Design a phase shift oscillator using operational amplifier to generate a 5 kHz signal. Assume the supply voltage is ± 12 V.

Solution:

Let us consider $C = 10$ nF. Therefore, using Equ.(16-5),

$$R = \frac{1}{2\pi f_o C \sqrt{6}} = \frac{1}{2\pi \times 5 \text{ kHz} \times 10 \text{ nF} \sqrt{6}} = 1.3 \text{ k [Ans.]}$$

$$\therefore R_f \geq 29R_3 \geq 29R \geq 29 \times 1.3 \text{ k} \geq 37.1 \text{ k, use standard } 39 \text{ k [Ans.]}$$

Alternately, let us consider $C = 1$ nF. Therefore, using Equ.(16-5),

$$R = \frac{1}{2\pi f_o C \sqrt{6}} = \frac{1}{2\pi \times 5 \text{ kHz} \times 1 \text{ nF} \sqrt{6}} = 13 \text{ k [Ans.]}$$

$$\therefore R_f \geq 29R_3 \geq 29R \geq 29 \times 13 \text{ k} \geq 377 \text{ k, use standard } 390 \text{ k [Ans.]}$$

Comments: I simulated both the circuits, and found the circuit functioning properly.

16.6 Wien Bridge Oscillator

Like the Wheatstone bridge, a Wien bridge is also consists of four arms. Wheatstone bridge uses four resistors, while the Wien bridge has a series RC in one arm and a parallel RC in another arm. Therefore, it is an AC bridge and balances at a particular frequency. A Wien bridge oscillator is shown in Fig.16-8.

This oscillator uses Wien bridge as the frequency selecting or feedback network. Wien bridge oscillator needs a non-inverting amplifier, that is, an amplifier with zero phase shift. Here, cascading two CE stages a non-inverting amplifier is made. The output of the amplifier is applied to the bridge at point A. This output voltage will be

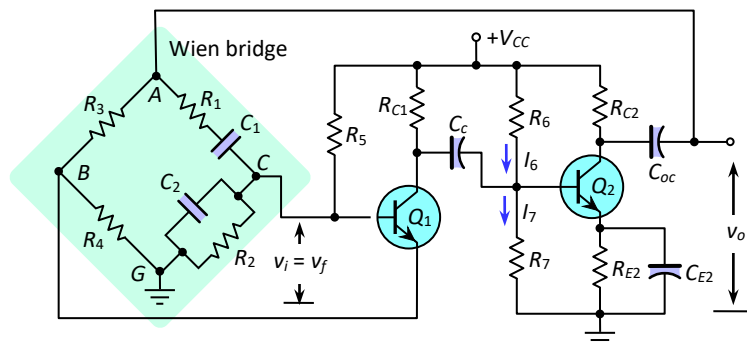


Fig.16-8: Wein bridge oscillator using BJT

divided by the two AC branches (branches with capacitors) of the bridge. Voltage at point C is applied to the base of amplifier and the voltage at point B is applied to the emitter of first transistor. Resistor R_3 and R_4 provides negative feedback to the amplifier, while the AC branches will provide positive feedback for oscillation. Here, the voltage applied to the base (voltage at point C) is the feedback voltage v_f . To achieve positive feedback the phase shift of v_f should be 0° . The Wien bridge will provide 0° phase shift at point C (i.e., for v_f) only at balanced condition and at that condition the feedback factor will be one third ($1/3$). The variation of β and phase shift with frequency is shown in Fig.16-9. Therefore, Wien bridge oscillator will produce the frequency, for

which the bridge is balanced. Normally the gain of the two stage amplifier will be very high, but negative feedback will set the voltage gain as required for sustained oscillation.

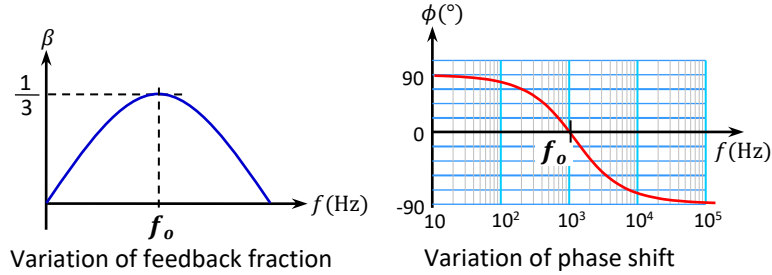


Fig.16-9: Variation of feedback signal with frequency

16.6.1 Analysis

The impedance of the branch, containing R_1 , and C_1 in series, will be

$$Z_s = R_1 + 1/j\omega C_1 \quad (16-11)$$

The impedance of the branch, containing R , and C in parallel, will be

$$\text{or,} \quad Z_p = R_2 \parallel 1/j\omega C_2 = \frac{R_2 \times 1/j\omega C_2}{R_2 + 1/j\omega C_2} = \frac{R_2}{1 + j\omega R_2 C_2} \quad (16-12)$$

Feedback factor can be calculated using voltage divider rule,

$$\beta = \frac{v_f}{v_o} = \frac{Z_p}{Z_s + Z_p}$$

Putting the values of Z_s , and Z_p from Equ.(16-11) and Equ.(16-12) we get,

$$\beta = \frac{R_2/(1 + j\omega R_2 C_2)}{R_1 + 1/j\omega C_1 + R_2/(1 + j\omega R_2 C_2)}$$

$$\text{or,} \quad \beta = \frac{1}{R_1(1 + j\omega R_2 C_2)/R_2 + (1 + j\omega R_2 C_2)/j\omega R_2 C_1 + 1}$$

$$\text{or,} \quad \beta = \frac{1}{R_1/R_2 + j\omega R_1 C_2 + 1/j\omega R_2 C_1 + C_2/C_1 + 1}$$

$$\text{or,} \quad \beta = \frac{1}{(R_1/R_2 + C_2/C_1 + 1) + j(\omega R_1 C_2 - 1/\omega R_2 C_1)} \quad (16-13)$$

To make $\angle\beta = 0$ at the frequency of oscillation ($\omega = \omega_o$), the imaginary part should be zero,

$$\therefore \quad \omega_o R_1 C_2 - 1/\omega_o R_2 C_2 = 0$$

$$\text{or,} \quad \omega_o^2 = \frac{1}{R_1 R_2 C_1 C_2}$$

$$\therefore \quad \omega_o = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$

$$\therefore \boxed{f_o = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}} \quad (16-14)$$

Let us consider $C_1 = C_2 = C$, and $R_1 = R_2 = R$.

$$\therefore \boxed{f_o = \frac{1}{2\pi RC}} \quad (16-15)$$

Considering the imaginary term zero, Equ.(16-13) becomes

$$\boxed{\beta = \frac{1}{(R/R + C/C + 1) + 0} = \frac{1}{3}} \quad (16-16)$$

In order to make $A_v \beta \geq 1$, the voltage gain of the amplifier should be $A_v \geq 3$.

The balance conditions of the Wien bridge is found in different books. So, here, I didn't analyze the bridge for balanced condition. A Wien bridge circuit goes to balanced state when the following two equations are satisfied.

$$\frac{R_3}{R_4} = \frac{R_1}{R_2} + \frac{C_2}{C_1}$$

$$\boxed{f_o = \frac{1}{2\pi RC}}$$

16.6.2 Design of BJT Wien Bridge Oscillator

Generally we design a Wien bridge oscillator for a particular frequency of oscillation for a given supply voltage. The design requirement is to find the components values. The two-stage amplifier should be designed with a voltage gain of 3. In this oscillator two types of feedback work. The positive feedback for oscillation is produced by the AC arm of the bridge, while the negative feedback is produced by the DC arm (R_3 and R_4) of the bridge. The output voltage is divided by R_3 and R_4 , and V_{R4} is applied to the emitter of Q_1 . As the overall phase shift of the two-stage amplifier is zero, the feedback signal applied to the emitter will work as negative feedback. To sustain oscillation the bridge must be balanced. To balance the bridge the following condition has to be satisfied.

$$\frac{R_3}{R_4} = \frac{R_1}{R_2} + \frac{C_2}{C_1}$$

Generally we select, $R_1 = R_2$, and $C_1 = C_2$.

$$\frac{R_3}{R_4} = \frac{R_1}{R_2} + \frac{C_2}{C_1} = 2$$

$$R_3 = 2R_4$$

Therefore, the value of negative feedback factor (β_{fn}) is,

$$\beta_{fn} = \frac{R_4}{R_3 + R_4} = \frac{R_4}{2R_4 + R_4} = \frac{1}{3}$$

We have to design the amplifier stages with gain as much as possible. However, the overall gain of the amplifier (A_v) will remain almost constant to 3 ($= 1/\beta_{fn}$) according to negative feedback theory. Equ.(16-16) gives the value of positive feedback $\beta = 1/3$. So the loop-gain will be $A_v\beta = 1$. But to get continuous oscillation, $A_v\beta$ should be a little greater than 1. This can be achieved by making R_3 a little greater than $2R_4$ that will decrease β_{fn} .

Finally, we have to select a suitable value of $C_1 = C_2 = C$ and from the required frequency of oscillation, value of $R = R_1 = R_2$ has to be calculated using Equ.(16-15). To understand the design process see Example 16-3.

Example 16-3

Design a Wien bridge oscillator to generate 132 kHz and 2 V amplitude. Assume the supply voltage is 12 V and the transistors have $h_{fe} = 200$.

Solution:

As the required amplitude is 2 V, let us assume $V_{CE1} = V_{CE2} = 3$ V, $V_{E1} = V_{E2} = 2$ V, and $I_{C1} = I_{C2} = 1$ mA.

$$\therefore R_4 = R_{E2} = \frac{2 \text{ V}}{1 \text{ mA}} = 2 \text{ k}$$

Therefore, R_3 should be a little greater than 4 k. Let it be 4.7 k.

$$R_{C1} = R_{C2} = \frac{V_{CC} - (V_E + V_{CE})}{1 \text{ mA}} = \frac{12 \text{ V} - (2 \text{ V} + 3 \text{ V})}{1 \text{ mA}} = 7 \text{ k, use } 6.8 \text{ k}$$

$$I_{B1} = I_{B2} = \frac{1 \text{ mA}}{200} = 5 \mu\text{A}$$

Let $I_7 = 0.1$ mA.

$$R_7 = \frac{V_{E2} + V_{BE2}}{I_7} = \frac{2 \text{ V} + 0.7 \text{ V}}{0.1 \text{ mA}} = 27 \text{ k,}$$

$$R_6 = \frac{V_{CC} - (V_{E2} + V_{BE2})}{I_7} = \frac{12 \text{ V} - 2.7 \text{ V}}{0.1 \text{ mA}} = 93 \text{ k, use standard } 91 \text{ k}$$

R_4 will work as the emitter resistance of Q_1 . Therefore,

$$I_{B1} = \frac{V_{CC} - V_{BE1}}{R_5 + h_{fe}R_4} \quad \text{or,} \quad R_5 + h_{fe}R_4 = \frac{V_{CC} - V_{BE1}}{I_{B1}}$$

$$\therefore R_5 = (12 \text{ V} - 0.7 \text{ V})/(5 \mu\text{A}) - 200 \times 2 \text{ k} = 1.66 \text{ M, use standard } 1.6 \text{ M}$$

Use all coupling and bypass capacitors of 1 μF .

Comments: As h_{fe} is very high, we can assume $I_6 = I_7$. According to feedback theory the overall gain of this two-stage amplifier will be independent of open-loop gain.



16.7 Wien Bridge Oscillator using Op-Amp

Fig.16-10 shows a Wien Bridge oscillator using operation amplifier. Here, R_4 is the feedback resistor and R_3 is the input resistor of the non-inverting amplifier. Same as BJT Wein bridge oscillator R_3 and R_4 provides negative feedback for the operational amplifier. The AC branch divides output voltage and produces v_f which is applied to the non-inverting terminal. The gain of the non-inverting amplifier is,

$$A_v = 1 + \frac{R_4}{R_3}$$

The value of feedback factor will be same as discussed for transistor Wien bridge oscillator. That is,

$$\beta = \frac{1}{(R_1/R_2 + C_2/C_1 + 1) + j(\omega R_1 C_2 - 1/\omega R_2 C_2)}$$

Therefore, the frequency of oscillation and the amplifier gain will be same as before,

$$f_o = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$$

Considering, $C_1 = C_2 = C$, and $R_1 = R_2 = R$.

$$f_o = \frac{1}{2\pi RC} \quad (16-17)$$

The closed-loop gain of the amplifier should be 3.

$$\therefore A_v = 1 + \frac{R_4}{R_3} = 3$$

$$\text{or,} \quad \frac{R_3}{R_4} = 2 \quad \text{or,} \quad R_3 = 2R_4 \quad (16-18)$$

As mentioned before, Wien bridge gets balanced when two equations are satisfied. These are,

$$\frac{R_3}{R_4} = \frac{R_1}{R_2} + \frac{C_2}{C_1} = \frac{R}{R} + \frac{C}{C} = 2$$

and,

$$f_o = \frac{1}{2\pi RC}$$

Therefore, for the balanced condition of the bridge, the amplifier can produce required gain for oscillation.

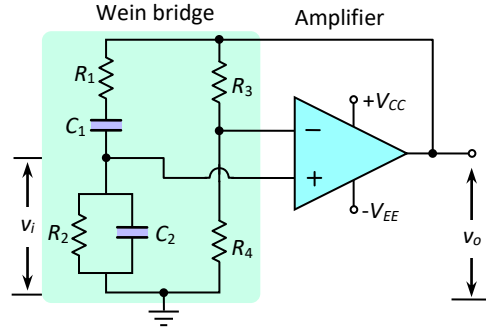


Fig.16-10: Wein bridge oscillator using op-amp

16.7.1 Design:

As before, the design requirement is mainly the frequency of oscillation. Generally the supply voltage is known. The maximum output voltage swing will be $\pm(\text{supply voltage} - 1)$. Alternately, if the output swing is given the supply voltage should be one volt larger. The design target is to select a suitable operational amplifier and components values. We know that $R_4 = 2R_3$. So, the maximum voltage across $R_3 + R_4$ will be $\pm(\text{supply voltage} - 1)$. Now, we have to consider a suitable current through the branches of the bridge. This current should be much larger than the bias circuit of operational amplifier. Considering a suitable value of C , the value of R can be calculated using the required frequency of oscillation using Equ.(16-7). Value of C should be much larger than the parasitic capacitance of Op-amp.

Example 16-4

Design a Wien bridge oscillator to produce a 200 kHz signal. The supply voltage of the operational amplifier is ± 6 V.

Solution:

As the supply voltage is ± 6 V, the peak-to-peak value of the output will be ± 5 V.

To produce a 200 kHz output the minimum SR of the Op-Amp should be [Using Equ.(.)],

$$SR = 2\pi f_p v_p = 2\pi(200 \text{ kHz}) \times 11 \text{ V} \\ = 6.28 \text{ V}/\mu\text{s}$$

The gain bandwidth product of the operational amplifier should be greater than $3 \times 200 \text{ kHz} = 600 \text{ kHz}$.

Now we have to select a suitable operational amplifier with sufficient slew rate and other requirements. Let us use LF353. Its slew rate is $13 \text{ V}/\mu\text{s}$ and gain bandwidth product is 4 MHz. The input bias current of this op-amp is 200 pA. Let the current through the branches of the bridge be 0.1 mA at the peak voltage.

$$\therefore R_3 + R_4 = \frac{5 \text{ V}}{0.1 \text{ mA}} = 50 \text{ k}$$

Using Equ.(16-18)

$$R_3 = 2R_4$$

$$\therefore R_3 = 50 \text{ k}/3 = 16.7 \text{ k, use standard } 16 \text{ k}$$

$$\therefore R_4 = 2 \times 16 \text{ k} = 32 \text{ k, use standard } 33 \text{ k}$$

Select $C_1 = C_2 = 1 \text{ nF}$

$$\text{Using Equ.(16-17), } R = \frac{1}{2\pi f_o C} = \frac{1}{2\pi \times (200 \times 10^3)(1 \times 10^{-9} \text{ F})} \approx 795.77 \Omega \\ \text{use } 750 \Omega + 43 \Omega = 793 \Omega$$

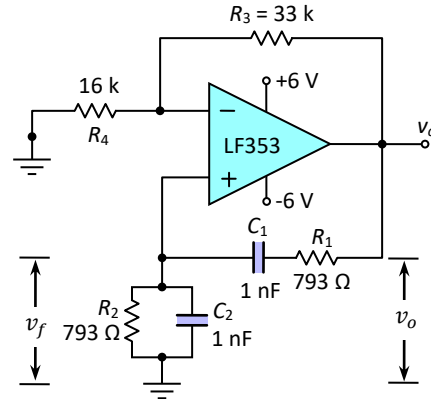


Fig.16-11: Wein bridge oscillator of Example 16-4

[the reactance of the AC branch of the bridge at 200 kHz will be $(793 + 793 + 793 \parallel 793) \Omega \approx 2 \text{ k}$. So, the current through this branch will be $5 \text{ V} / 2 \text{ k} = 2.5 \text{ mA}$. This current is quite larger than the bias current.]

Answer: The designed circuit is shown in Fig.16-11.

Comments: If 2.5 mA seems to be high, we can decrease the value of C say 0.5 nF. Lower value of C will result in higher reactance as well as higher values for R . So the current will be lower.

16.8 Colpitts Oscillator

Colpitts oscillator is an LC oscillator that generates medium to high frequencies. The frequency range is 20 kHz to 300 MHz. Fig.16-12 shows the circuit diagram of a Colpitts oscillator. This circuit consists of a CE amplifier with voltage divider bias and a frequency selecting network. The frequency selecting network is made up of an inductor L and capacitors C_1 and C_2 . This parallel combination of capacitors and inductor is also called **tank circuit**. The amplifier amplifies the signal and produces 180° phase shift and the feedback network produces another 180° phase shift. C_{ic} and C_{oc} are the coupling capacitors and C_E is the emitter bypass capacitor. For the frequency of oscillation these capacitors will act as short circuits. Note that instead of resistor an inductor (coil) is used in the collector. This coil is called **RFC** (Radio Frequency Choke) that provides the necessary DC load resistance (internal resistance) for collector and also prevents AC signal from being short through the voltage source (V_{CC}). As the DC resistance of RFC is small, it provides a wide range to set the operating point and reduces DC power loss.

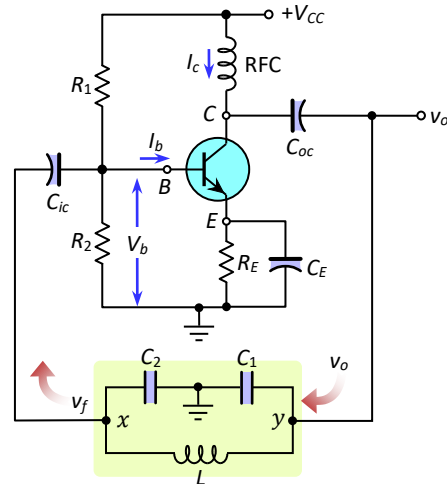


Fig.16-12: Colpitts oscillator

16.8.1 Circuit operation

To understand the operation of LC oscillator, first we will discuss the operation of an LC tank circuit. An LC tank circuit is parallel combination of L and C . If a supply voltage is connected momentarily to this tank circuit, the capacitor will immediately be fully charged. If the power supply is disconnected from the tank circuit, a damped oscillation will be produced. The operation procedure is illustrated in Fig.16-13. Now the capacitor will discharge through the inductor and the current will gradually increase and the energy will be stored in the inductor as magnetic energy. When the capacitor is fully discharged, the inductor will try to oppose the change in the current. Actually then, the inductor will work as a source and helps the current to flow in the same direction. In the figure this current is represented as i_m . Thus the capacitor is charged in opposite polarity. When all of the energy of the inductor is used up, the current (i_m) stops. In the meantime the capacitor is fully charged with opposite polarity, that is all of the magnetic energy has been converted to electrostatic energy again. Now the capacitor starts to discharge and produces current

through the inductor in opposite direction. Again the electrostatic energy is converted into magnetic energy. If there were no loss of energy, this energy conversion process would have been

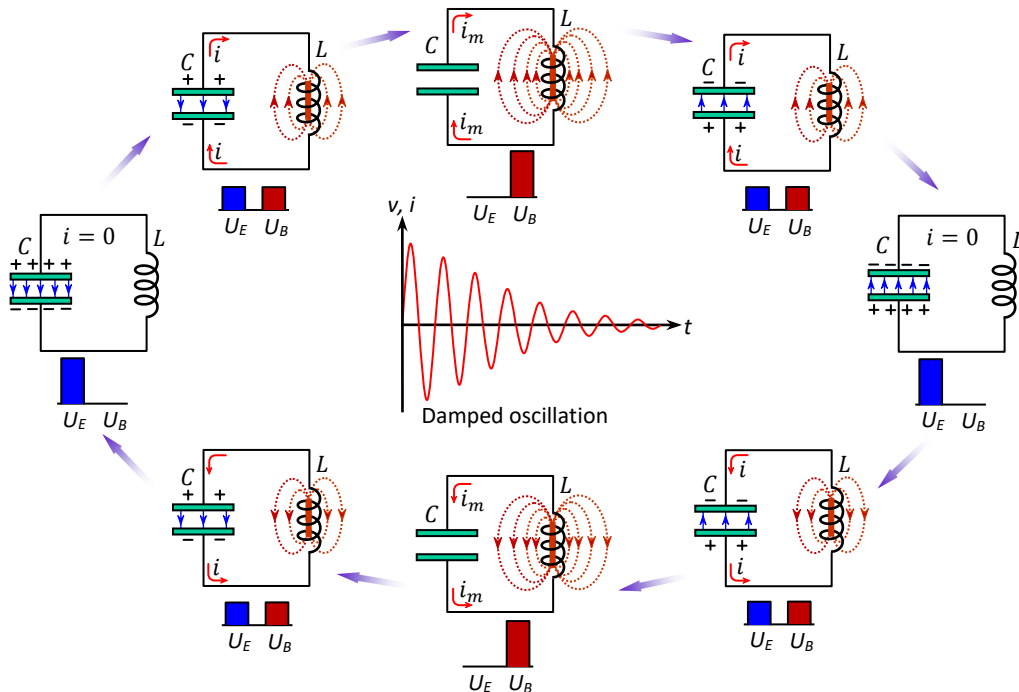


Fig.16-13: Oscillation mechanism of a LC tank circuit

continued infinitely and a pure sinusoidal oscillation was produced. However, due to the resistive loss, mainly in the internal resistance of the inductor, the amplitude of the oscillation gradually decrease, and a damped sinusoidal waveform (voltage and current) is produced. In this way any tank circuit produces a damped oscillation. The frequency of oscillation is the parallel resonant frequency of the tank circuit at which the capacitive reactance (X_C) becomes equal to the inductive reactance (X_L).

When the supply voltage of the Colpitts oscillator is switched ON, collector current starts to flow and capacitors C_1 and C_2 are charged up. These capacitors and the inductor L work as a tank circuit and damped oscillation is produced. The capacitors C_1 and C_2 act as a voltage divider network for the generated AC signal. The portions of the AC signal dropped across C_1 are applied to amplifier's input and an amplified output is developed at the output (collector). The amount of feedback depends on the ratio of C_1 and C_2 (that is C_1/C_2). If C_2 increases, its impedance decreases, hence feedback voltage decreases and vice versa. The total phase shift around the loop is 360° . Voltages at point x , which is the feedback voltage (v_f) is 180° out of phase from the voltage at point y which is the output voltage of the amplifier. Thus the feedback circuit produces 180° phase shift. Another 180° phase shift is produced by the CE amplifier. Therefore, the energy supplied by the amplifier to the tank circuit is in phase with the generated oscillations in it. The amplifier provides sufficient amount of energy that is lost in the tank circuit. Hence, a continuous undamped oscillation is produced. The frequency of oscillation will be the resonant frequency of the parallel LC circuit. At this frequency the reactance of equivalent capacitance ($C_1 \parallel C_2$) will be

equal to the reactance of the inductor. Output is generally taken as shown in Fig.16-12, but the output can also be taken using a secondary coil with the inductor.

16.9 Hartley Oscillator

Hartley oscillator is another common LC oscillator. Typical operating frequency of Hartley oscillator is from 20KHz to 20MHz and it can be realized using BJT, FET or op-amps. Fig.16-14 shows the circuit diagram of a Hartley oscillator using a CE amplifier. The amplifier is made using a BJT with voltage divider bias. Capacitors C_{ic} and C_{oc} are used as coupling capacitors and C_E is the emitter bypass capacitor. For frequency of oscillation these capacitors will work as almost short circuits. Here also, instead of R_C a RFC is used in the collector to provide AC load and to block the signal from being shorted. The frequency determining network is made up of capacitor C and inductors L_1 and L_2 . Inductors L_1 and L_2 are wound in a single core, so they are mutually coupled and work as an auto transformer. AC signal developed across L_2 is applied to the input of amplifier through C_{ic} . Feedback between output and input is accomplished through transformer action. However, we can use two separate inductors, and then mutual inductance will be zero. The oscillation developed in the tank circuit will be divided between the coils. As the right terminal of L_2 is grounded and the left terminal is connected to input, the feedback signal will be 180° phase shifted. CE amplifier will produce another 180° phase shift. Thus total phase shift around the loop will be 360° . This ensures positive feedback which is the essential requirement for oscillation. Radio frequency choke (RFC) provides AC load for the collector and also prevent AC signal from being shorted via power supply. The reactance of RFC should be higher than that of L_1 so that it does not affect the operation of tank circuit. The amount of feedback depends on the values of L_2 and L_1 (that is L_2/L_1). If L_2 increases, its impedance will also increase, hence feedback voltage will increase and vice versa.

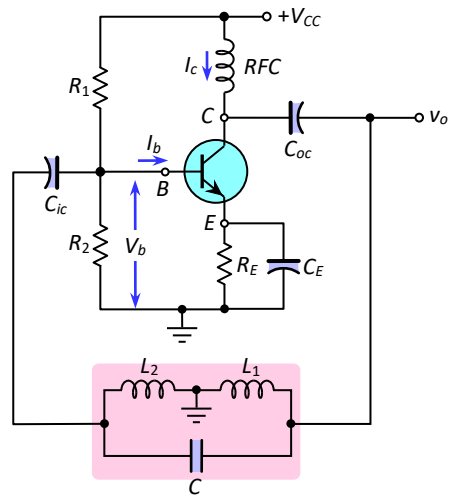


Fig.16-14: Hartley oscillator using BJT

16.9.1 Circuit Operation

When the power supply is switched ON, collector current starts to rise, and charges the capacitor (C). When C is fully charged, it discharges through coils L_1 and L_2 . Thus damped oscillations are set up in this tank circuit as described before. The oscillations developed across L_2 are applied to amplifier's input (base emitter junction). These oscillations are amplified and appear at the output (collector terminal). Feedback of energy from collector-emitter circuit to the base-emitter circuit is accomplished by means of mutual inductance between L_1 and L_2 or simply by the division of voltage. The amplifier amplifies the feedback voltage and continuously supplies energy to the tank circuit to overcome the losses occurring in it. So, continuous undamped oscillations are obtained at

the output. Like the Colpitts oscillator, the Hartley oscillator also produces frequency at which the parallel LC circuit resonates.

16.9.2 General Analysis of Colpitts and Hartley Oscillator

A complete analysis of Colpitts and Hartley oscillators using bipolar transistors is complicated. To simplify the analysis we make the following assumptions:

- (i) The biasing resistors R_1 and R_2 are so large that they have no effect on the AC operation of the circuit.
- (ii) Since output resistance r_o of transistor is very high, its effect can be neglected.
- (iii) Since h_{re} of transistor is usually very small, $h_{re}v_o$ will be neglected.

With these assumptions, the general equivalent circuit for both Colpitts and Hartley oscillators is drawn in Fig.16-15.

The impedance Z_2 and the input resistance of the transistor h_{ie} work in parallel. Let the equivalent impedance be Z'_2

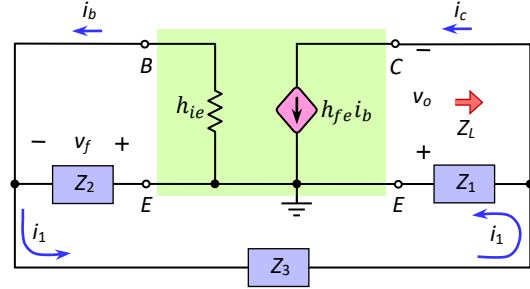


Fig.16-15: General equivalent circuit of Colpitts and Hartley oscillators

$$\therefore Z'_2 = Z_2 \parallel h_{ie} = \frac{Z_2 h_{ie}}{Z_2 + h_{ie}} \quad (16-19)$$

Now the load impedance between collector and emitter terminals is,

$$Z_L = Z_1 \parallel (Z_3 + Z'_2)$$

$$\text{or, } Z_L = \frac{Z_1 (Z_3 + Z'_2)}{Z_1 + Z_3 + Z'_2} \quad (16-20)$$

The voltage gain of the amplifier is,

$$A_v = -\frac{h_{fe} Z_L}{h_{ie}}$$

Output voltage can be calculated as,

$$v_o = i_1 Z_1 = i_1 (Z'_2 + Z_3)$$

The feedback voltage to the input (base-emitter terminals) is,

$$v_f = i_1 Z'_2$$

Therefore, the feedback factor is,

$$\beta = \frac{v_f}{v_o} = \frac{i_1 Z'_2}{i_1 (Z'_2 + Z_3)} = \frac{Z'_2}{(Z'_2 + Z_3)}$$

The loop-gain of the amplifier with feedback is,

$$A_v\beta = -\frac{h_{fe}Z_L}{h_{ie}} \times \frac{Z'_2}{(Z'_2 + Z_3)}$$

To sustain the oscillation, we have to fulfill the Barkhausen criteria

That is,

$$A_v\beta = 1 \quad \text{or,} \quad -\frac{h_{fe}Z_L}{h_{ie}} \times \frac{Z'_2}{(Z'_2 + Z_3)} = 1$$

Substituting the value of Z_L from Equ.(16-20) into the above equation, we get,

$$-\frac{h_{fe}}{h_{ie}} \times \frac{Z_1(Z_3 + Z'_2)}{Z_1 + Z_3 + Z'_2} \times \frac{Z'_2}{(Z'_2 + Z_3)} = 1$$

or,

$$\frac{h_{fe}}{h_{ie}} \times \frac{Z_1Z'_2}{Z_1 + Z_3 + Z'_2} = -1$$

Putting the value of Z'_2 from Equ.(16-19) into the above equation,

$$\frac{h_{fe}}{h_{ie}} \times \frac{Z_1[Z_2h_{ie}/(Z_2 + h_{ie})]}{Z_1 + Z_3 + Z_2h_{ie}/(Z_2 + h_{ie})} = -1$$

or,

$$\frac{Z_1Z_2h_{fe}}{Z_1(Z_2 + h_{ie}) + Z_3(Z_2 + h_{ie}) + Z_2h_{ie}} = -1$$

or,

$$Z_1Z_2h_{fe} = -Z_1(Z_2 + h_{ie}) - Z_3(Z_2 + h_{ie}) - Z_2h_{ie}$$

or,

$$Z_1Z_2h_{fe} = -Z_1Z_2 - Z_1h_{ie} - Z_2Z_3 - Z_3h_{ie} - Z_2h_{ie}$$

or,

$$\boxed{h_{ie}(Z_1 + Z_2 + Z_3) + Z_1Z_2(1 + h_{fe}) + Z_2Z_3 = 0} \quad (16-21)$$

This is the general conditional expression for both Colpitts and Hartley oscillator.

16.9.3 Analysis of Colpitts Oscillator

As shown in Fig.16-12, two capacitors C_1 and C_2 are used in place of Z_1 and Z_2 and an inductor, L is used in place of Z_3 . Therefore,

$$Z_1 = 1/j\omega C_1, \quad Z_2 = 1/j\omega C_2, \quad \text{and} \quad Z_3 = j\omega L$$

Substituting these values in Equ.(16-21), we get,

$$\text{or,} \quad h_{ie} \left(\frac{1}{j\omega C_1} + \frac{1}{j\omega C_2} + j\omega L \right) + \left(\frac{1}{j\omega C_1} \right) \left(\frac{1}{j\omega C_2} \right) (1 + h_{fe}) + \left(\frac{1}{j\omega C_2} \right) j\omega L = 0$$

$$\text{or,} \quad -jh_{ie} \left(\frac{1}{\omega C_1} + \frac{1}{\omega C_2} - \omega L \right) - \frac{1}{\omega^2 C_1 C_2} (1 + h_{fe}) + \frac{L}{C_2} = 0 \quad (16-22)$$

To make the left-hand side of the above equation zero, both the imaginary part and real part should be zero and this will occur at the frequency of oscillation, that is, $\omega = \omega_o$. Equating the imaginary part to zero, we get,

$$-jh_{ie} \left(\frac{1}{\omega_o C_1} + \frac{1}{\omega_o C_2} - \omega_o L \right) = 0$$

$$\text{or,} \quad \frac{1}{C_1} + \frac{1}{C_2} - \omega_o^2 L = 0$$

$$\text{or,} \quad \omega_o^2 = \frac{C_1 + C_2}{C_1 C_2 L} \quad (16-23)$$

$$\text{or,} \quad \omega_o = \sqrt{\frac{C_1 + C_2}{C_1 C_2 L}} = \sqrt{\frac{1}{C_T L}} \quad (16-24)$$

where, $C_T = C_1 || C_2 = C_1 C_2 / (C_1 + C_2)$.

$$\text{or,} \quad \boxed{f_o = \frac{1}{2\pi\sqrt{C_T L}}} \quad (16-25)$$

This is the expression for the frequency of oscillation.

Equating the real part of Equ.(16-22) to zero, we get,

$$-\frac{1}{\omega_o^2 C_1 C_2} (1 + h_{fe}) + \frac{L}{C_2} = 0$$

$$\text{or,} \quad (1 + h_{fe}) = \omega_o^2 C_1 L$$

Putting the value of ω_o^2 from Equ.(16-23) into the above equation, we get,

$$\text{or,} \quad (1 + h_{fe}) = \left(\frac{C_1 + C_2}{C_1 C_2 L} \right) C_1 L = \frac{C_1 + C_2}{C_2} = \frac{C_1}{C_2} + 1$$

$$\text{or,} \quad h_{fe} = \frac{C_1}{C_2} = \frac{\text{Capcitanace connected to collector}}{\text{Capcitanace connected to base}} \quad (16-26)$$

Equ.(16-26) gives the minimum value of h_{fe} for the sustained oscillations.

16.9.4 Analysis of Hartley Oscillator

As shown in Fig.16-14, for Hartley oscillator two inductors (L_1 and L_2) are used in place of Z_1 and Z_2 and a capacitor, C is used in place of Z_3 . Let us consider that the resistance of the inductor is so small that we can neglect it and M is the value of mutual inductance between the inductors (coils). Therefore,

$$Z_1 = j\omega L_1 + j\omega M, \quad Z_2 = j\omega L_2 + j\omega M, \quad \text{and} \quad Z_3 = 1/j\omega C$$

Substituting these values in Equ.(16-21), we get,

$$\begin{aligned} & h_{ie}(j\omega L_1 + j\omega M + j\omega L_2 + j\omega M + 1/j\omega C) \\ & + (j\omega L_1 + j\omega M)(j\omega L_2 + j\omega M)(1 + h_{fe}) + (j\omega L_2 + j\omega M)1/j\omega C = 0 \end{aligned}$$

$$\text{or,} \quad j\omega h_{ie}(L_1 + L_2 + 2M - 1/\omega^2 C) - \omega^2(L_1 + M)(L_2 + M)(1 + h_{fe}) + (L_2 + M)/C = 0 \quad (16-27)$$

At the frequency of oscillation, ω_o , the imaginary part and the real part of this equation will be zero. Equating the imaginary part to zero, we get,

$$\text{or,} \quad j\omega h_{ie}(L_1 + L_2 + 2M - 1/\omega_o^2 C) = 0$$

$$\text{or,} \quad \frac{1}{\omega_o^2 C} = L_1 + L_2 + 2M$$

$$\text{or,} \quad \omega_o^2 = \frac{1}{(L_1 + L_2 + 2M)C} \quad (16-28)$$

$$\text{or,} \quad \omega_o = \frac{1}{\sqrt{(L_1 + L_2 + 2M)C}} \quad (16-29)$$

$$\text{or,} \quad \boxed{f_o = \frac{1}{2\pi\sqrt{(L_1 + L_2 + 2M)C}}} \quad (16-30)$$

This equation gives the value of frequency of oscillation of Hartley oscillator.

Equating the real part to zero, we get

$$\text{or,} \quad -\omega_o^2(L_1 + M)(L_2 + M)(1 + h_{fe}) + (L_2 + M)/C = 0$$

$$\text{or,} \quad \omega_o^2(L_1 + M)(1 + h_{fe}) = \frac{1}{C}$$

$$\text{or,} \quad 1 + h_{fe} = \frac{1}{\omega_o^2(L_1 + M)C}$$

Putting the value of ω_o^2 from Equ.(16-28) into the above equation, we get,

$$\text{or,} \quad 1 + h_{fe} = \frac{(L_1 + L_2 + 2M)C}{(L_1 + M)C} = 1 + \frac{L_2 + M}{L_1 + M}$$

$$\text{or,} \quad h_{fe} = \frac{L_2 + M}{L_1 + M} = \frac{\text{Inductance connected to base}}{\text{Inductance connected to collector}} \quad (16-31)$$

Equ.(16-) gives the minimum value of h_{fe} to get the continuous oscillations from the oscillator.

16.9.5 Alternative Analysis

For both Colpitts and Hartley oscillators, the frequency of oscillation is the resonant frequency of the tank circuit. Resonance occurs when the reactance of two branches are equal. So for Colpitts oscillator,

$$|X_L| = |X_{CT}|$$

$$\text{or,} \quad \omega_o L = 1/\omega_o C_T$$

$$\text{or,} \quad \omega_o^2 = 1/LC_T$$

$$\text{or,} \quad \omega_o = 1/\sqrt{LC_T}$$

$$\therefore \quad f_o = \frac{1}{2\pi\sqrt{LC_T}}$$

$$\text{where,} \quad C_T = C_1 \parallel C_2 = C_1 C_2 / (C_1 + C_2)$$

At the frequency of oscillation,

$$X_L = X_{CT} \text{ or, } j\omega_o L = 1/j\omega_o C_T = (C_2 + C_1)/C_1 C_2 j\omega_o$$

$$\text{or, } j\omega_o L = 1/j\omega_o C_1 + 1/j\omega_o C_2$$

$$\text{or, } X_L = X_{C1} + X_{C2}$$

$$\therefore X_{C2} = X_L - X_{C1} \quad (16-32)$$

From the Colpitts oscillator circuit, we find that the amplifier's output is divided between L and C_2 . So, the feedback factor will be,

$$\beta = \frac{-jX_{C2}}{jX_L - jX_{C2}} = -\frac{X_{C2}}{X_L - X_{C2}}$$

$$\text{Using Equ. (16-32)} \quad \beta = -\frac{X_{C2}}{X_L - X_L + X_{C1}} = -\frac{X_{C2}}{X_{C1}}$$

$$\text{or, } \beta = -\frac{1/\omega C_2}{1/\omega C_1} = -\frac{C_1}{C_2} \quad (16-33)$$

The above equation gives the value of feedback factor. The negative sign indicates 180° phase shift.

From Barkhausen criteria,

$$A_v \beta \geq 1$$

$$\therefore A_v \geq \frac{1}{\beta} = -\frac{C_2}{C_1} \quad (16-34)$$

The above equation gives the minimum gain requirement of the amplifier.

Hartley oscillator:

At the frequency of oscillation,

$$|X_C| = |X_{LT}|$$

$$\text{or, } 1/\omega_o C = \omega_o L_T$$

$$\text{or, } \omega_o^2 = 1/L_T C$$

$$\text{or, } \omega_o = 1/\sqrt{L_T C}$$

$$\therefore f_o = \frac{1}{2\pi\sqrt{L_T C}} \quad (16-35)$$

$$\text{where, } L_T = L_1 + L_2$$

At the frequency of oscillation,

$$X_C = X_{LT} \quad \text{or, } 1/j\omega_o C = j\omega_o L_T = j\omega_o (L_1 + L_2)$$

$$\text{or, } 1/j\omega_o C = j\omega_o L_1 + j\omega_o L_2$$

$$\text{or, } X_C = X_{L1} + X_{L2}$$

$$\therefore X_{L2} = X_C - X_{L1} \quad (16-36)$$

From Fig.16-14, we find that the amplifier's output is divided between C and L_2 . So, the feedback factor will be,

$$\beta = \frac{jX_{L2}}{-jX_C + jX_{L2}} = -\frac{X_{L2}}{X_C - X_{L2}}$$

$$\text{Using Equ.(16-32)} \quad \beta = -\frac{X_{L2}}{X_C - X_C + X_{L1}} = -\frac{X_{L2}}{X_{L1}}$$

$$\text{or,} \quad \beta = -\frac{\omega L_2}{\omega L_1} = -\frac{L_2}{L_1} \quad (16-37)$$

The above equation gives the value of feedback factor. The negative sign indicates 180° phase shift.

From Barkhausen criteria,

$$A_v \beta \geq 1$$

$$\therefore A_v \geq \frac{1}{\beta} = -\frac{L_1}{L_2} \quad (16-38)$$

The above equation gives the minimum gain requirement of the amplifier.

Another calculation of β

The impedance of the tank circuit will be very high at resonant frequency (theoretically ∞). So the output current will not affect its operation. As shown in Fig.16-16, if the current developed in the tank circuit, i , flows anti-clock-wise, voltage across C_1 will be positive and the voltage across C_2 will be negative. The output voltage will be iX_{C1} and the feedback voltage will be, $-iX_{C2}$. Therefore, the feedback factor will be,

$$\beta = \frac{v_f}{v_o} = -\frac{iX_{C2}}{iX_{C1}} = -\frac{1/j\omega C_2}{1/j\omega C_1} = -\frac{C_1}{C_2}$$

Similarly for Hartley oscillator,

$$\beta = \frac{v_f}{v_o} = -\frac{iX_{L2}}{iX_{L1}} = -\frac{j\omega L_2}{j\omega L_1} = -\frac{L_2}{L_1}$$

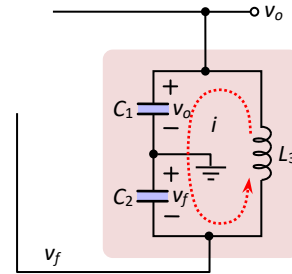


Fig.16-16: Process of 180° phase shift of v_f

16.10 Design of Colpitts and Hartley Oscillators

Equ.(16-33) shows that for Colpitts oscillator C_1 will be smaller than C_2 to make $\beta < 1$. We can select a suitable value for C_1 depending on the required frequency of oscillation. Its value should be low- but much larger than the stray capacitances of BJT (15 pF to 20 pF), otherwise the resonant frequency of the tank circuit will be overridden by the stray capacitance. The values of C_2 should be larger than C_1 . We can select $C_2 = 5C_1$ to $20C_1$ or any suitable value depending on the required frequency of oscillation or the value of inductance. Moreover, we should make X_{C2} much larger than the output impedance of the amplifier. Once the values of C_1 and C_2 are fixed, we have to

determine the value of $C_T = C_1 C_2 / (C_1 + C_2)$. The values of L can be calculated plugging the value of frequency and C_T into Equ.(16-25). The input impedance of CE amplifier is $R_B \parallel h_{ie} \approx h_{ie}$. This input impedance should be large enough to avoid the loading effect on the tank circuit. Feedback factor will be $|\beta| = C_1/C_2 = 1/5$ to $1/20$. Now we have to design the CE amplifier with voltage gain 5 to 20. The RFC coil has to be selected to produce sufficient reactance at the frequency of oscillation to produce required gain of the amplifier. In fact, we can use resistor in place of RFC (See example 15-5). To get higher input impedance of the amplifier, we can keep the emitter resistor or some portion of the emitter resistance unbypassed.

To design of Harley oscillator, we can use same concept. Here, L_1 will be larger than L_2 . We can select $L_1 = 5L_2$ to $20L_2$. The gain of the amplifier should be same as Colpitts oscillator. The input impedance of the amplifier should be high, and the output impedance should be low compared to the impedance of the tank circuit.

Example 16-5

Design a Colpitts oscillator to generate 250 kHz signal. Assume that the supply voltage is 12 V, and the transistor has $h_{fe} = 200$.

Solution:

Let $C_1 = 4.7$ nF (a standard value) and $C_2 = 47$ nF. Therefore,

$$C_T = \frac{C_1 C_2}{C_1 + C_2} = \frac{4.7 \text{ nF} \times 47 \text{ nF}}{4.7 \text{ nF} + 47 \text{ nF}} = 4.27 \text{ nF}$$

$$\text{Using Equ.(16-25)} \quad L = \frac{1}{(2\pi f_o)^2 C_T} = \frac{1}{(2\pi \times 250 \text{ kHz})^2 \times 4.27 \text{ nF}} = 95 \text{ } \mu\text{H}$$

Let $I_C = 1 \text{ mA} \approx I_E$, and $V_E = 3 \text{ V}$

$$\therefore R_E = \frac{V_E}{I_E} = \frac{3 \text{ V}}{1 \text{ mA}} = 3 \text{ k}$$

Let the current through R_2 be $I_2 = I_C/10 = 0.1 \text{ mA}$

$$\therefore R_2 = \frac{V_E + V_{BE}}{I_2} = \frac{3 \text{ V} + 0.7 \text{ V}}{0.1 \text{ mA}} = 37 \text{ k, use standard } 36 \text{ k}$$

$$\therefore R_1 = \frac{V_{CC} - V_{R2}}{I_2} = \frac{12 \text{ V} - 3.7 \text{ V}}{0.1 \text{ mA}} = 83 \text{ k, use standard } 82 \text{ k}$$

$$\text{Using Equ.(16-)} \quad r_e = \frac{26 \text{ mV}}{I_E} \approx \frac{26 \text{ mV}}{1 \text{ mA}} = 26 \text{ } \Omega$$

Feedback factor will be,

$$\text{Using Equ.(16-33)} \quad |\beta| = \frac{C_1}{C_2} \approx \frac{4.7 \text{ nF}}{47 \text{ nF}} = 0.1$$

The voltage gain of the amplifier should be greater than $1/\beta = 1/0.1 = 10$.

Using Equ.(16-),

$$A_v = \frac{R_C}{r_e} \quad \text{or,} \quad R_C = 10 \times 26 \, \Omega = 260 \, \Omega$$

To make the voltage gain a little greater we can set $R_C = 300 \, \Omega$. But to get a good amplitude of the oscillation we have to make R_C even higher than $300 \, \Omega$.

Using Equ.(9-7),

$$C_E = \frac{1}{2\pi f_L h_{ib}} = \frac{1}{2\pi f_L r_e} = \frac{1}{2\pi(250 \, \text{kHz}) \times 26 \, \Omega} = 24 \, \text{nF}$$

This is the required minimum capacitance. Let us use all coupling and bypass capacitors ten times C_2 .

$$C_E = C_{ic} = C_{oc} = C_o = 470 \, \text{nF}$$

Results: The designed circuit has been simulated using Ltspice and is given with the output signal in Fig.15.

Comments: From the simulation, we see that the signal's amplitude gradually increases and then becomes fixed.

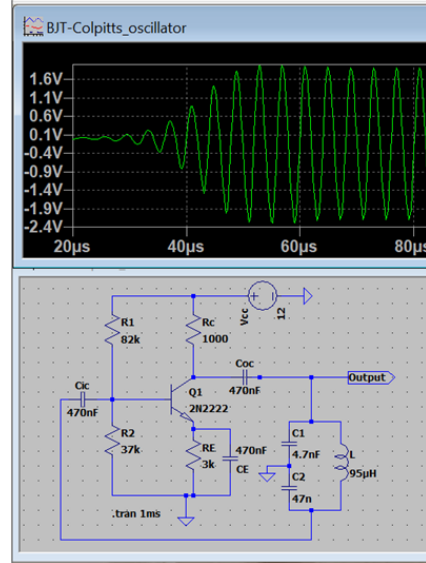


Fig.16-17: Colpitts oscillator of Example 15-5 (simulated in Ltspice)

Example 16-6

Design a Hartley oscillator to generate 300 kHz signal. Assume that the supply voltage is 12 V, and the transistor has $h_{fe} = 200$.

Solution:

Let $L_2 = 0.5 \, \mu\text{H}$, and $L_1 = 10L_2 = 5 \, \mu\text{H}$

$$L_T = L_1 + L_2 = 5.5 \, \mu\text{H}.$$

Using Equ.(16-35),

$$C = \frac{1}{(2\pi f_o)^2 L_T} = \frac{1}{(2\pi \times 300 \, \text{kHz})^2 \times 5.5 \, \mu\text{H}} = 51 \, \text{nF}$$

Here the feedback factor is $\beta = L_2/L_1 = 0.5 \, \mu\text{H}/5 \, \mu\text{H} = 0.1$.

Therefore, the amplifier can be designed exactly in the same way as Example 15-5. All coupling and bypass capacitor should be used ten times greater than the capacitance of the tank circuit.

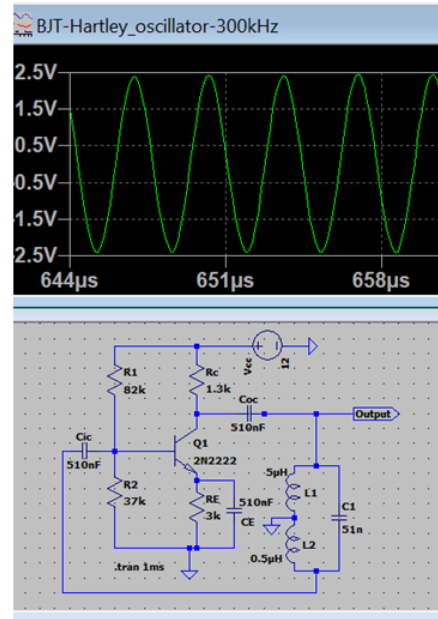


Fig.16-18: Hartley oscillator of Example 15-6 (simulated in Ltspice)

Results: The designed circuit has been simulated using Ltspice and is given with the output signal in Fig.16-18.

Comments: Here also, the signal's amplitude gradually increases, but not shown in the simulation result.

16.11 Colpitts Oscillators using Operational Amplifier

Fig.16-19 shows a Colpitts oscillator using operational amplifier. Here the op-amp is used as an inverting amplifier that produces 180° phase shift. Another 180° phase shift is produced by the tank circuit as described in BJT oscillator. Here also the expression for the frequency of oscillation will be,

$$f_o = \frac{1}{2\pi\sqrt{LC_T}} \quad \text{Same as Equ.(16-25)}$$

where, $C_T = C_1 \parallel C_2 = C_1 C_2 / (C_1 + C_2)$

The value of feedback factor will be,

$$\beta = -\frac{C_1}{C_2}$$

To make $A_v \beta = 1$, the gain of the amplifier has to be,

$$A_v = -\frac{C_2}{C_1} = -\frac{R_f}{R_1} \quad (16-39)$$

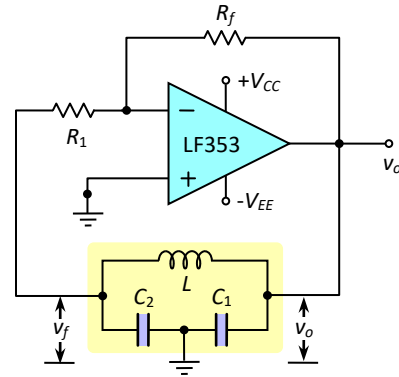


Fig.16-19: Colpitts oscillator using op-amp

16.12 Hartley Oscillators using Operational Amplifier

Fig.16-20 shows a Hartley oscillator using operational amplifier. Like the Colpitts oscillator the inverting amplifier produces 180° phase shift and the required gain. Another 180° phase shift is produced by the tank circuit. Here also the expression for the frequency of oscillation will be,

$$f_o = \frac{1}{2\pi\sqrt{(L_1 + L_2 + 2M)C}} \quad \text{Same as Equ.(16-25)}$$

The value of feedback factor will be,

$$\beta = -\frac{L_2}{L_1}$$

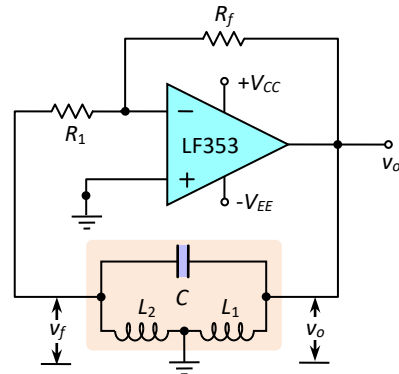


Fig.16-20: Hartley oscillator using op-amp

The gain of the amplifier has to be,

$$A_v = -\frac{L_1}{L_2} = -\frac{R_f}{R_1} \quad (16-40)$$

16.12.1 Design of Colpitts and Hartley Oscillator Using Op-Amp

In Colpitts oscillator C_2 will be greater than C_1 . Value of C_1 should be small but much larger than stray capacitance (15 to 20 pF). We can select $C_2 = 5C_1$ to $20C_1$ or any suitable value for a given value of inductance. However, we should make X_{C1} much larger than the output impedance of op-amp. Value of L can be calculated plugging the value of frequency and C_T into Equ.(16-25). The input impedance of inverting amplifier is R_1 . Therefore, the value of R_1 should be large enough to avoid the loading effect on the tank circuit. Feedback factor can be set to 1/5 to 1/20. Now the inverting amplifier has to be designed with voltage gain 5 to 20, that is, $R_f = 5R_1$ to $20R_1$.

To design Hartley oscillator we can use same concept. Here, L_1 will larger than L_2 . We can select $L_1 = 5L_2$ to $20L_2$. The gain of the amplifier should be 5 to 20. Here also X_{L1} should be much larger than output impedance of the op-amp, and the value of R_1 should be much larger than X_{L2} .

Example 16-7

Design a Colpitts oscillator using operational amplifier to produce a 100 kHz signal. The supply voltage of the operational amplifier is ± 12 V.

Solution:

Let $C_1 = 10$ nF and $C_2 = 10C_1 = 100$ nF.

$$\therefore C_T = C_1 \parallel C_2 = 100 \text{ nF} \parallel 100 \text{ nF} = 9.1 \text{ nF}$$

$$\text{Using Equ.(16-25), } L = \frac{1}{(2\pi f_o)^2 C_T} = \frac{1}{(2\pi \times 100 \text{ kHz})^2 \times 9.1 \text{ nF}} \approx 2.8 \text{ } \mu\text{H} \quad 280 \text{ } \mu\text{H}$$

Let $R_1 = 10$ k

$$\text{Using Equ.(16-39), } \frac{R_f}{R_1} = \frac{C_2}{C_1} = \frac{100 \text{ nF}}{10 \text{ nF}} = 10$$

$$\text{or, } R_f = 10R_1 = 100 \text{ k}$$

Answer: $C_1 = 10$ nF, $C_2 = 100$ nF, $L = 280 \text{ } \mu\text{H}$, $R_1 = 10$ k, and $R_f = 100$ k

Comments: We could have designed the oscillator considering $C_1 = 1$ nF.

16.13 Clapp Oscillator

The Clapp oscillator design was published by James Kilton Clapp in 1948^[2]. In fact, the Clapp oscillator is an advanced version of Colpitts oscillator. As shown in Fig.16-21, the circuit differs from the Colpitts oscillator only in one respect: it contains an additional capacitor (C_3) connected in series with the inductor. The addition of capacitor (C_3) improves the frequency stability and eliminates the effect of transistor parameters and stray capacitances.

Operation of Clapp oscillator circuit is the same as that of Colpitts oscillator. The amplifier provides voltage gain and 180° phase shift. The output voltage is divided across C_1 and C_2 , and the voltage across C_2 is feedback to the amplifier. So the feedback ratio is $\beta = X_{C2}/X_{C1} = C_1/C_2$ (same as Colpitts oscillator). As the upper part of C_2 is grounded, the feedback signal is 180° phase shifted from output signal. The amplifier produces another 180° phase shift. Thus, the total phase shift is 360° . The gain of the amplifier should be $A_v \geq 1/\beta = C_2/C_1$. When damped oscillation is produced in the tank circuit all of the three capacitances work in series. Therefore, the equivalent capacitance will determine the frequency of oscillator. From Equ.(16-25), we can write,

$$f_o = \frac{1}{2\pi\sqrt{LC_T}} \quad (16-41)$$

where,

$$C_T = 1/\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)$$

Generally, the value of C_3 is much smaller than both C_1 and C_2 . Therefore, we can assume $C_T \approx C_3$, that is, the oscillation frequency depends only on C_3 ,

$$f_o = \frac{1}{2\pi\sqrt{LC_3}} \quad (16-42)$$

Clapp oscillator has many advantages over the Colpitts oscillator. It is easier to vary capacitance than inductance. In Colpitts oscillator, we can vary the frequency by changing either C_1 or C_2 but, the problem is the feedback factor will also be changed. In the Clapp oscillator, the frequency depends mainly on C_3 . So we can easily generate variable frequency by Clapp oscillator. As C_3 is made very small, the value of required inductance is large. Therefore, the high reactance of this inductor isolates C_3 from the parasitic capacitance of the amplifier. Thus, oscillation frequency of Clapp oscillator is not affected by the device parameters. We know that the Q -factor can be represented as $Q = \frac{1}{R}\sqrt{L/C}$. As the ratio L/C_3 is very high, Clapp oscillator has a very high Q -factor and hence very good frequency selectivity.

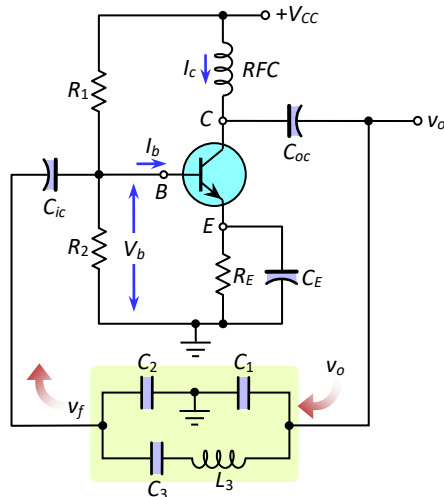


Fig.16-21: Clapp oscillator

Example 16-8

Design a Clapp oscillator to generate a signal of variable frequency 200 kHz to 500 kHz. Assume $V_{CC} = 12$ V and $h_{fe} = 200$.

Solution:

Let $C_1 = 47$ nF (a standard value) and $C_2 = 470$ nF. Therefore, the gain of the amplifier should be little greater than 10. We can design the amplifier exactly in the same method of Example 15-5.

Let $C_3 = 1 \text{ nF}$. The value of L_3 can be determined using Equ.(16-42).

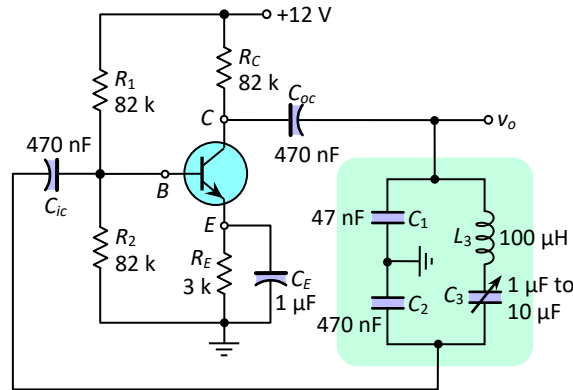


Fig.16-22: Clapp oscillator of Example 15-7

$$L = \frac{1}{(2\pi f_o)^2 C_3} = \frac{1}{(2\pi \times 500 \text{ kHz})^2 \times 1 \text{ nF}} \approx 101 \text{ } \mu\text{H}, \text{ use standard } 100 \text{ } \mu\text{H}$$

Now considering $L_3 = 100 \text{ } \mu\text{H}$, the highest value of C_3 will be found lowest frequency. Again using Equ.(16-42),

$$C_3 = \frac{1}{(2\pi f_o)^2 L_3} = \frac{1}{(2\pi \times 200 \text{ kHz})^2 \times 100 \text{ } \mu\text{H}} \approx 6.33 \text{ nF}$$

Let us use a trimmer with variable capacitance 1 nF to 10 nF in place of C_3 .

As the frequency of oscillation is high, let us use $C_{ic} = C_{oc} = 470 \text{ nF}$ and $C_E = 1 \text{ } \mu\text{F}$.

Answer: The designed circuit is shown in Fig.16-22

Comments: Clapp oscillator is suitable for generating variable frequency. We could have calculated the values of coupling and bypass capacitors, but the above assumption will work.

16.14 Tuned Collector Oscillator

Fig.16-23 shows the circuit diagram of tuned collector oscillator using CE amplifier. It contains a tuned circuit $L_1 C_1$ in the collector and hence it is called so. The combination of L_1 and C_1 forms the oscillatory circuit (tank circuit) to set the frequency of oscillation. Here C_1 may be a variable capacitor and L_1 forms the primary winding of a step down transformer. A feedback coil L_2 in the base circuit is magnetically coupled (with mutual inductance M) to coil L_1 of the tank circuit. Since transistor is connected in CE configuration, it provides a phase shift of 180° between its input and output. Another phase shift of 180° is provided by the transformer. In this way a phase shift of 360° appears between output and input voltages resulting in a positive feedback. The DC operating

point for the transistor is established by supply voltage V_{CC} emitter resistance R_E and potential divider arrangement (R_1 and R_2). C_{ic} and C_{oc} and the coupling capacitors and C_E is the emitter bypass capacitor.

Circuit operation: When the circuit is power up, collector current starts increasing and charges the capacitor C_1 . When capacitor C_1 is fully charged, it discharges through coil L_1 setting up natural oscillations in the tank circuit. These natural oscillations induce a small voltage across L_2 by mutual induction. This is the feedback voltage v_f . The value of v_f depends upon the number of turns in L_2 and coupling between L_1 and L_2 . The voltage across L_2 is applied between base and emitter and thus causes corresponding variation in base-current. The variations in I_b are amplified β times and appear in collector circuit. A part of the amplified energy is used in overcoming the losses occurring in tank circuit and the balance is radiated out in the form of electromagnetic waves.

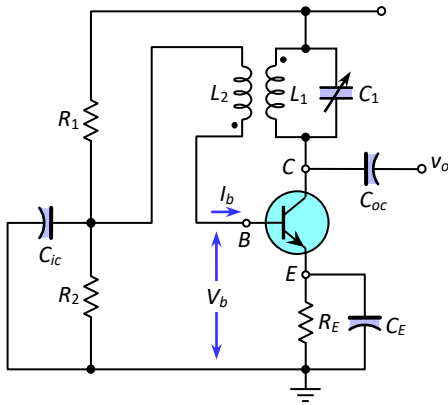


Fig.16-23: Tuned collector oscillator

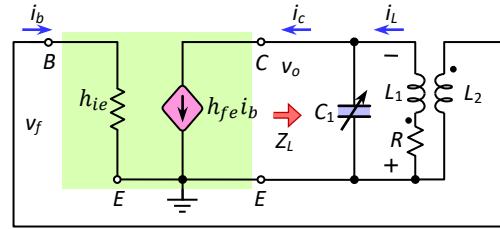


Fig.16-24: Equivalent circuit of tuned collector oscillator

Circuit Analysis

Considering $R_B = R_1 || R_2 \gg h_{ie}$, the equivalent circuit of the oscillator is drawn in Fig.(16-24).

From this circuit we find that the voltage gain of the amplifier is,

$$A_v = -\frac{h_{fe}Z_L}{h_{ie}} \quad (16-43)$$

Here, Z_L is the equivalent impedance at the output and its value will be,

$$Z_L = \left(\frac{1}{j\omega C_1} \right) || (R + j\omega L_1) = \frac{1/j\omega C_1 (R + j\omega L_1)}{1/j\omega C_1 + (R + j\omega L_1)}$$

or,

$$Z_L = \frac{R + j\omega L_1}{1 + j\omega C_1(R + j\omega L_1)} = \frac{R + j\omega L_1}{1 + j\omega C_1 R - \omega^2 C_1 L_1}$$

$\therefore$

$$Z_L = \frac{R + j\omega L_1}{(1 - \omega^2 C_1 L_1) + j\omega C_1 R}$$

Putting this value into Equ.(16-43) we get,

$$A_v = -\frac{h_{fe}}{h_{ie}} \times \frac{R + j\omega L_1}{(1 - \omega^2 C_1 L_1) + j\omega C_1 R} \quad (16-44)$$

From Fig.16-24, if i_L be current through L_1 , output voltage will be,

$$v_o = -i_L(R + j\omega L_1)$$

Let the mutual inductance between L_1 and L_2 be M . Due to this mutual inductance, current i_L in the primary coil will also induce a voltage in the secondary coil L_2 . This voltage will be applied to the input of the amplifier. So this will be the feedback voltage (v_f). According to the dot notation of the transformer v_f will be 180° out of phase from v_o . Value of v_f will be,

$$v_f = i_L j\omega M$$

Therefore, the feedback factor will be,

$$\beta = \frac{v_f}{v_o} = \frac{i_L j\omega M}{-i_L(R + j\omega L_1)} = -\frac{j\omega M}{(R + j\omega L_1)} \quad (16-45)$$

Multiplying Equ.(16-44) and Equ.(16-45), the loop-gain of the amplifier will be,

$$A_v \beta = \frac{h_{fe}}{h_{ie}} \times \frac{R + j\omega L_1}{(1 - \omega^2 C_1 L_1) + j\omega C_1 R} \times \frac{j\omega M}{(R + j\omega L_1)} = \frac{j\omega h_{fe} M}{h_{ie}[1 - \omega^2 C_1 L_1 + j\omega C_1 R]}$$

According to the Barkhausen criteria the magnitude of $A_v \beta$ should be unity.

$$\begin{aligned} \frac{j\omega h_{fe} M}{h_{ie}[1 - \omega^2 C_1 L_1 + j\omega C_1 R]} &= 1 \\ j\omega h_{fe} M &= h_{ie} - \omega^2 h_{ie} C_1 L_1 + j\omega h_{ie} C_1 R \\ h_{ie} - \omega^2 h_{ie} C_1 L_1 + j\omega(h_{ie} C_1 R - h_{fe} M) &= 0 \end{aligned}$$

At the frequency of oscillation the real part and the imaginary part of this equation will separately zero. Equating the real part to zero, we get,

$$h_{ie} - \omega_o^2 h_{ie} C_1 L_1 = 0$$

$$\omega_o^2 h_{ie} C_1 L_1 = h_{ie}$$

$$\omega_o^2 = \frac{h_{ie}}{h_{ie} C_1 L_1} = \frac{1}{C_1 L_1}$$

$$\omega_o = \frac{1}{\sqrt{C_1 L_1}}$$

$$\boxed{f_o = \frac{1}{2\pi\sqrt{L_1 C_1}}}$$

(16-46)

This is the expression for the frequency of oscillation of the oscillator.

Again equating the imaginary part to zero, we get,

$$j\omega(h_{ie} C_1 R - h_{fe} M) = 0$$

$$h_{ie}C_1R - h_{fe}M = 0$$

$$M = \frac{h_{ie}C_1}{h_{fe}L_2} \quad (16-47)$$

Equ.(16-47) gives the minimum value of mutual inductance for which the oscillator will work.

Example 16-9

For the tuned collector oscillator of Fig.16-, determine the frequency of oscillation and the minimum required gain of the amplifier.

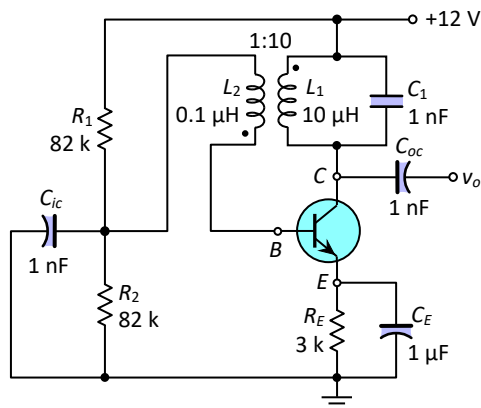


Fig.16-25: Tuned collector oscillator of Example 15-8

Solution:

From the figure, the values of $L_1 = 10 \mu\text{H}$, $L_2 = 0.1 \mu\text{H}$, $C_1 = 1 \text{ nF}$, and the turn ratio is $N_1:N_2 = 10:1$.

Using Equ.(16-46)
$$f_o = \frac{1}{2\pi\sqrt{L_1C_1}} = \frac{1}{2\pi\sqrt{10 \mu\text{H} \times 1 \text{ nF}}} \approx 1.6 \text{ MHz [Ans.]}$$

The feedback factor will be,

$$\beta = N_2:N_1 = 1/10$$

Therefore, the gain of the amplifier should be greater than 10 [Ans.].

Comments:

16.15 Tuned Base Oscillator

Tune base oscillator is another type of transistor based LC oscillator where the tuned circuit (tank circuit) is placed between the base and ground of the transistor. The tuned base oscillator is also called *Armstrong oscillator* after the name of its inventor. The tuned circuit essentially consists of a

transformer and capacitor connected in parallel. The circuit diagram of a typical tuned base oscillator is shown in Fig.16-26.

16.15.1 Working principle of tuned base oscillator

When power is applied, the transistor Q_1 starts conducting and its collector current starts to rise. From the circuit diagram, we can see that this collector current actually flows through secondary coil L_2 . This rise in current through the secondary induces some voltage across the primary coil L_1 by virtue of mutual induction. The secondary coil L_2 is also called the *Tickler Coil*. The induced voltage across L_1 charges the capacitor C_1 . The energy stored in the coil L_1 as electromagnetic field is transferred into the capacitor as electrostatic field. Now the capacitor starts discharging through L_1 . This energy transfer from C_1 and L_1 produces a damped oscillation in the tank circuit as described before. These oscillations are coupled to the base of the transistor through capacitor C_{ic} and it appears in amplified form at the collector of the transistor. A portion of the collector voltage is inductively coupled to the tank circuit by the secondary coil L_2 for compensating the losses in the tank circuit. The feedback fraction will be $N_1:N_2$ and can be adjusted by varying the turn ratio. The winding directions of L_1 and L_2 are so selected that the transformer produces 180° phase shift between the primary voltage and secondary voltage. The CE amplifier produces another 180° phase shift. The amplifier must have a gain greater than $1/\beta = N_2/N_1$. The frequency of oscillations of the tuned base oscillator will be same as the tuned collector oscillator, that is,

$$f_o = \frac{1}{2\pi\sqrt{L_1 C_1}}$$

The main drawback of tuned-base oscillator circuit is that, due to the low base-emitter resistance, which appears in shunt with the tank circuit, it gets loaded. This reduces its Q -factor which in turn causes drift in oscillator frequency.

16.16 Crystal Oscillator

One of the most important features of any oscillator is its *frequency stability*, or in other words its ability to provide a constant frequency output under varying load conditions. Some of the factors that affect the frequency stability of an oscillator generally include: variations in temperature, variations in the load, as well as changes to its DC power supply voltage, etc. Frequency stability of the output signal can be greatly improved by proper selection of the components used for the resonant feedback circuit, including the amplifier. But there is a limit to the stability that can be achieved from normal LC and RC tank circuits. However, high level of stability can be obtained by using crystals in oscillators.

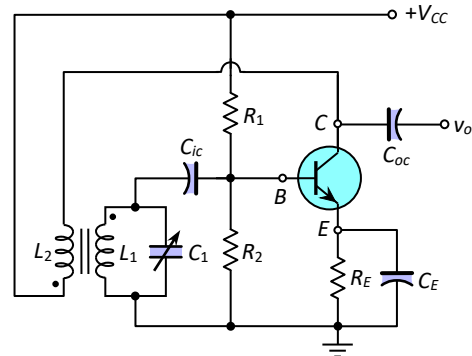


Fig.16-26: Tuned base oscillator

16.16.1 Crystal

Some crystals like- quartz, Rochelle salt, tourmaline, corundum etc. show *piezoelectric effect*. When mechanical stress is applied to a piece of such crystals, electric charges are accumulated on the faces of this crystal. This phenomenon is called *piezoelectric effect*. As shown in Fig.16-27, due to the applied stress the crystal structure is deformed, and net positive and negative charges are produced in two opposite faces. On the contrary, if an alternating voltage is applied to two opposite faces of a piece of crystal, it mechanically vibrates. The vibration of crystal with applied voltage is called *reverse piezoelectric effect*. The vibration is the so-called thickness-shear or face-shear vibration, in which both surfaces of the crystal plate shift w.r.t. each other as shown in Fig.16-28. There are also crystals which resonate by bending like a tuning fork: among others the small 32.768 kHz crystals [Fig.16-30(b)] in watches work like this. These crystals have this particular frequency of 32.768 kHz, because if we pass this frequency through 15 Flip-Flops, we get 1 Hz ($32768 \text{ Hz} \div 2^{15} = 1 \text{ Hz}$).

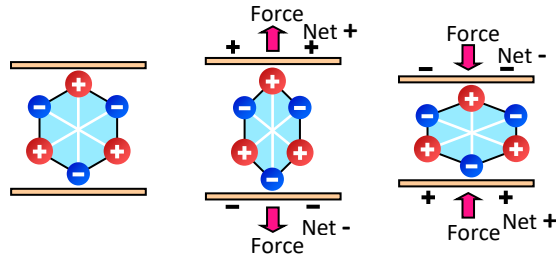


Fig.16-27: Piezoelectric effect

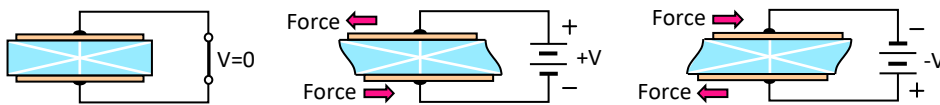


Fig.16-28: Reverse piezoelectric effect (thickness shear)

Rachelle salt shows maximum piezoelectric effect, but they are fragile. On the other hand, tourmaline is mechanically rugged but shows least piezoelectric effect. Quartz crystal is rugged and shows good piezoelectric effect. For this reason quartz is mostly used as a piezoelectric material. Quartz is a hard, crystalline mineral composed of silicon and oxygen atoms (SiO_4). As natural quartz is costly to mine, most of the quartz used for crystal fabrication today is of the *cultured* or *synthetic* variety. To make the device (resonator), quartz crystal is cut in different orientation and thickness as shown in Fig.16-29. Different cutting angle has different names and different properties. Once cut and shaped, the crystal cannot be used at any other frequency. The frequency of oscillation is determined by the type of cut, cutting angle, dimensions of the crystal piece, and construction of the electrodes. The crystals characteristic or characteristic frequency is inversely proportional to its physical thickness between the two metallic surfaces.

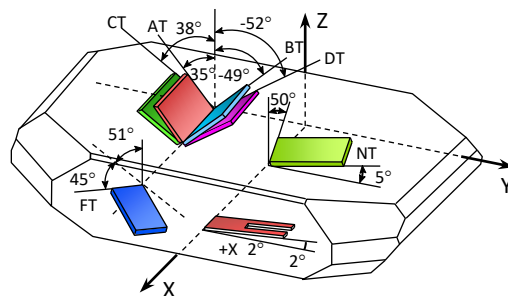


Fig.16-29: Different cutting angles of quartz crystal

After cutting, the piece of crystal is placed between two conducting plates that works as the external electrodes of the crystal. As electrodes, low resistance materials like silver, gold or aluminum are used. Fig.16-30 some crystal resonators commonly used in electronics.

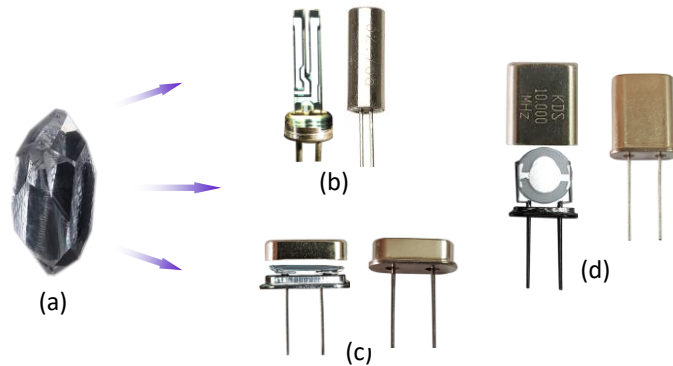


Fig.16-30: Photographs of different crystal resonators: (a) Bulk crystal, (b) Resonator used in watch, (c) Small resonator used in microprocessor, and (d) Big resonator used in microprocessor

16.16.2 Quartz Crystal Equivalent Model and Response

A mechanically vibrating crystal can be represented by an equivalent electrical circuit consisting of a low value motional resistance, R_s ($10\ \Omega$ to $150\ \Omega$), a large motional inductance L_s (few mH to H) and small motional capacitance C_s (2 fF to 50 fF) as shown in Fig.16-31(b). As the crystal is placed in between two metal plates, they form a capacitance which is represented by C_m (mounting capacitance). Its value ranges from 0.5 pF to 10 pF. When a crystal vibrates, its **inertia** is equivalent to L_s , **stiffness** is equivalent to C_s and **frictional loss** is equivalent to R_s . The quality factor (Q) of an inductor/crystal is defined as $2\pi L/R$. The value of R in a crystal is so small that it can give quality factor in the range of 10^4 to 10^6 , while the quality factor of a very good inductor is 10^3 . The stability of an oscillator depends on the quality factor. As the quality factor of a crystal is very large, crystal oscillator provides very stable oscillation frequency. This is the main advantage of a crystal oscillator.

The equivalent electrical circuit for the quartz crystal shows a series RLC circuit, which represents the mechanical vibrations of the crystal, in parallel with a capacitance, C_m which represents the electrical connections to the crystal. Quartz crystal oscillators tend to operate either in their **series resonance** or **parallel resonance**. In series resonance, C_s resonates with inductance, L_s at a particular frequency and gives the minimum impedance ($Z_p = R$). On the other hand, at parallel resonance, C_m resonates with L_s at another particular frequency and gives the maximum impedance. The series resonant frequency (f_s) is lower than the parallel resonant frequency (f_p), but the difference is not more than 1%. When properly drive, a crystal can generate overtone frequencies, which are the odd harmonics of f_p . The impedance vs. frequency graph of a typical crystal is shown in Fig.16-31(c).

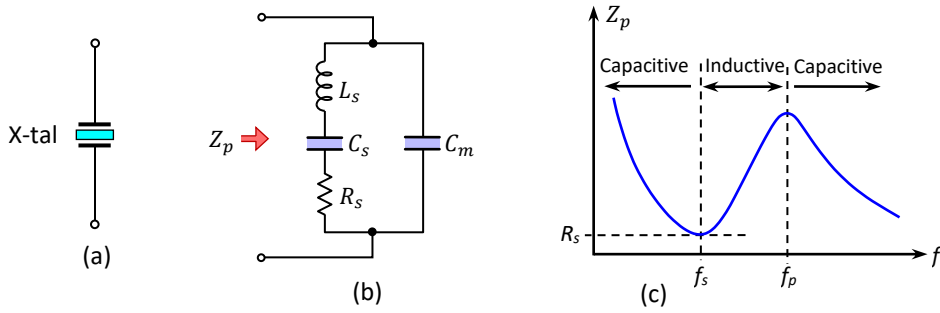


Fig.16-31: (a) Symbol of crystal, (b) Equivalent circuit, and (c) Characteristics

From the equivalent model of the crystal [Fig.16-31(b)], its impedance can be represented as,

$$Z_p = Z_s || Z_m = (j\omega L_s + 1/j\omega C_s + R_s) || (1/j\omega C_m)$$

$$\because R_s \text{ is very small} \quad Z_p = \left(j\omega L_s + \frac{1}{j\omega C_s} \right) || \left(\frac{1}{j\omega C_m} \right)$$

$$\text{or,} \quad Z_p = \frac{(1 - \omega^2 L_s C_s)}{j\omega C_s} || \left(\frac{1}{j\omega C_s} \right) = \frac{[(1 - \omega^2 L_s C_s)/j\omega C_s](1/j\omega C_m)}{(1 - \omega^2 L_s C_s)/j\omega C_s + 1/j\omega C_m}$$

$$\text{or,} \quad Z_p = \frac{(1 - \omega^2 L_s C_s)}{j\omega C_m(1 - \omega^2 L_s C_s) + j\omega C_s}$$

$$\text{or,} \quad Z_p = \left(-\frac{j}{\omega} \right) \frac{(1 - \omega^2 L_s C_s)}{C_m(1 - \omega^2 L_s C_s) + C_s}$$

$$\text{or,} \quad Z_p = \left(\frac{j}{\omega} \right) \times \frac{(\omega^2 L_s C_s - 1)}{C_s + C_m - \omega^2 L_s C_s C_m} \quad (16-48)$$

From this equation we find that Z_p depends on frequency (ω). At zero frequency (DC), $j/\omega = \infty$ so, the impedance will be infinity. For AC signal, Z_p has two special values. In one case $Z_p = 0$, that will be found when $\omega^2 L_s C_s - 1 = 0$. Actually this is the condition of series resonance. Therefore, at series resonance,

$$\omega_s^2 L_s C_s - 1 = 0$$

$$\text{or,} \quad \omega_s^2 = \frac{1}{L_s C_s}$$

$$\text{or,} \quad f_s = \frac{1}{2\pi\sqrt{L_s C_s}} \quad (16-49)$$

Series resonance occurs at lower frequency at which the reactance of mounting capacitance is very large. So it is absent in Equ.(16-49).

In another case, Z_p will be infinity for the condition $C_s + C_m - \omega^2 L_s C_s C_m = 0$. This is the condition of parallel resonance. Therefore, at parallel resonance,

$$C_s + C_m - \omega_p^2 L_s C_s C_m = 0$$

or,

$$\omega_p^2 = \frac{C_s + C_m}{L_s C_s C_m} = \frac{1}{L_s C_T}$$

where, $C_T = C_s C_m / (C_s + C_m) = C_s \parallel C_m$.

or,

$$f_p = \frac{1}{2\pi\sqrt{L_s C_T}} \quad (16-50)$$

In parallel resonance, the circulating current flows through the loop. So, it encounters both C_s and C_m in series. The equivalent of two series capacitors is $C_s \parallel C_m$. Therefore, this equivalent capacitance appears in Equ.(16-50).

16.16.3 Quality Factor or Q-factor:

Quality factor or Q -factor (Q) of a crystal or (coil) is defined as the ratio of the maximum energy stored to the maximum energy dissipated per cycle of the resonant frequency. It can be shown that

$$Q = 2\pi f \frac{\text{maximum energy stored}}{\text{maximum energy dissipated}} = \frac{X_L}{R} = \frac{2\pi f L_s}{R} \quad (16-51)$$

Since the value L_s of a crystal is very large and R is very small, the quality factor of a crystal is very large.

Example 16-10

A quartz crystal has $R_s = 5 \Omega$, $C_s = 25 \text{ fF}$, $L_s = 50 \text{ mH}$, and the mounting capacitance $C_m = 5 \text{ pF}$. Calculate the series and parallel resonant frequencies of the crystal and its Q -factors at these resonant frequencies.

Solution:

The series oscillating frequency can be calculated using Equ.(16-49).

$$f_s = \frac{1}{2\pi\sqrt{L_s C_s}} = \frac{1}{2\pi\sqrt{(50 \times 10^{-3})(25 \times 10^{-15})}} \approx 4502 \text{ kHz [Ans.]}$$

The parallel resonant frequency can be calculated using Equ.(16-50).

The equivalent capacitance is,

$$C_T = \frac{C_s C_m}{C_s + C_m} = \frac{25 \text{ fF} \times 5 \text{ pF}}{25 \text{ fF} + 5 \text{ pF}} = 24.876 \text{ fF}$$

$$\therefore f_p = \frac{1}{2\pi\sqrt{L_s C_T}} = \frac{1}{2\pi\sqrt{(50 \times 10^{-3})(24.876 \times 10^{-15})}} = 4513 \text{ kHz [Ans.]}$$

Using Equ.(16-51)

$$Q_s = \frac{2\pi f_s L_s}{R_s} = \frac{2\pi(4502 \text{ kHz}) \times (50 \text{ mH})}{5 \Omega} = 282.9 \times 10^3 \text{ [Ans.]}$$

$$Q_p = \frac{2\pi f_p L_s}{R_s} = \frac{2\pi(4513 \text{ kHz}) \times (50 \text{ mH})}{5 \Omega} = 283.6 \times 10^3 \text{ [Ans.]}$$

Comments: Generally the parallel resonant frequency is few kHz larger than the series resonant frequency.

16.17 Colpitts Oscillator with Crystal

In Colpitts oscillator the capacitors and the inductor determine the frequency of oscillation. As the Q -factor of a coil is low, the frequency of Colpitts oscillator drifts as described before. In place of inductor, if a crystal is used more stable frequency is obtained. In this case, the equivalent inductance of the crystal forms the tank circuit and determines the frequency of oscillation. We can replace the inductor of Fig.16-13 by a crystal of desired frequency. However, it is possible to design Colpitts oscillator using a common collector amplifier as shown in Fig.16-32(a). In this circuit, the crystal works as an inductor and produces a frequency little greater than its series resonant frequency (f_s). A damped oscillation is produced in the tank circuit consisting of the crystal's inductance and the capacitors C_1 and C_2 . The output voltage is applied across C_2 and the feedback voltage is taken from across capacitors $C_1 + C_2$. Look at the polarity of the output voltage and the feedback voltage. They are in same polarities. So there is no phase shift in the feedback circuit. As the CC amplifier produces no phase shift, the total phase shift of the oscillator is 0° .

The gain of a common collector amplifier is $R_E/(r_e + R_E)$, which is less than unity. Here, the required gain for oscillation is provided by the tank circuit itself. We know that at resonance, $X_L = X_{C1} + X_{C2}$. As shown in Fig.16-32(b), the output voltage will produce two currents in the tank circuit. One current (i_2) flows through $-jX_{C2}$, and the other current (i_1) flows through the other branch containing $-jX_{C1}$, jX_{Ls} and R_s . As the input impedance of CC amplifier is very large, i_b is considered zero. Now we can calculate v_o and v_f using i_1 as

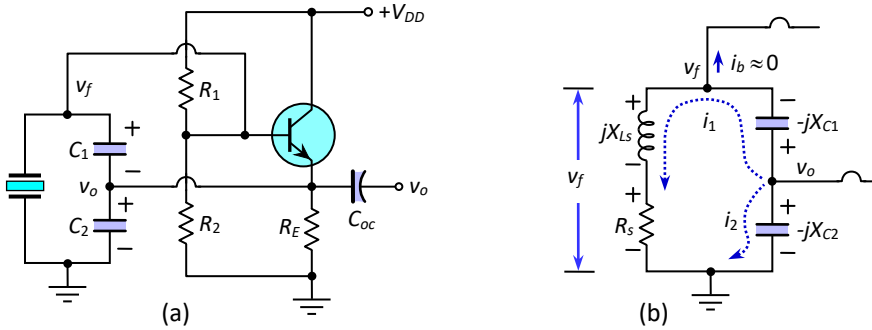


Fig.16-32: (a) Crystal Colpitts oscillator using CC amplifier, (b) Current direction in the tank circuit

Output voltage,

$$v_o = i_1(-jX_{C1} + jX_{Ls} + R_s)$$

Feedback voltage,

$$v_f = i_1(jX_{Ls} + R_s)$$

So, feedback factor,

$$\beta = \frac{v_f}{v_o} = \frac{i_1(jX_{Ls} + R_s)}{i_1(-jX_{C1} + jX_{Ls} + R_s)}$$

$$\because X_{Ls} = X_{C1} + X_{C2}$$

$$\beta = \frac{jX_{C1} + jX_{C2} + R_s}{-jX_{C1} + jX_{C1} + jX_{C2} + R_s}$$

$\because R_s$ is very small

$$\beta = \frac{jX_{C1} + jX_{C2}}{+jX_{C2}} = 1 + \frac{X_{C1}}{X_{C2}} = 1 + \frac{1/j\omega C_1}{1/j\omega C_2}$$

$\therefore$

$$\beta = 1 + \frac{C_2}{C_1} \quad (16-52)$$

We can find this value alternately. From Fig.16-32(b), the voltage across C_1 and C_2 is the feedback voltage, v_f . Similarly the voltage across C_2 is the output voltage, v_o . Here, C_1 and C_2 forms a voltage divider network. According to the voltage divider rule we can write,

$$v_o = \frac{v_f X_{C2}}{jX_{C1} + jX_{C2}}$$

$$\beta = \frac{v_f}{v_o} = \frac{X_{C1} + X_{C2}}{X_{C2}} = 1 + \frac{X_{C1}}{X_{C2}} = 1 + \frac{C_2}{C_1}$$

From this expression, we find that β increases with the increase of C_2 and decrease of C_1 . But larger value of C_2 will decrease its reactance that works in parallel with R_E . So the gain of the amplifier will also decrease. Therefore, we have to increase β by decreasing the value of C_1 .

The actual gain of the amplifier will be,

$$A_v = \frac{R_E \parallel X_{C2}}{r_e + R_E \parallel X_{C2}}$$

To satisfy the Barkhausen criteria, $A_v \beta$ should be greater than or equal to 1. Therefore,

$$A_v \beta = \frac{R_E \parallel X_{C2}}{r_e + R_E \parallel X_{C2}} \left(1 + \frac{C_2}{C_1} \right) \geq 1$$

$$\frac{R_E \parallel X_{C2}}{r_e + R_E \parallel X_{C2}} \geq \frac{C_1}{C_1 + C_2} \quad (16-53)$$

16.18 Crystal Oscillator Using Common Base Amplifier

Fig.16-33 shows a crystal oscillator using common base amplifier. It is also a Colpitts oscillator. Here, the crystal works as an inductor and forms a tank circuit with C_1 and C_2 . The resonant frequency of this tank circuit is the fundamental frequency. The exact frequency of oscillation will be determined by the external capacitance called load capacitance. The load (or total) capacitance for this circuit will be,

$$C_T = C_1 \parallel C_2 + C_s$$

Here, C_s is the stray capacitance and its value can be 2 to 5 pF.

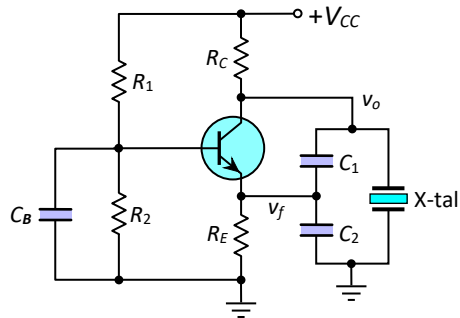


Fig.16-33: Crystal oscillator using CB amplifier

The crystal will oscillate with a frequency lower than the parallel resonant frequency. Thus, it will work as an inductive reactance [as shown in Fig.16-31(c)].

The amplifier is a common base amplifier with the base grounded for AC signals. Common base amplifier has no phase shift but high voltage gain. The output of the amplifier is taken from transistor's collector and applied to the tank circuit. The voltage across C_2 is the feedback voltage which is in phase with the output. That is the feedback voltage (v_f) has no phase shift. So the total phase shift of the circuit is 0° . Using voltage divider rule we can calculate the value of v_f as,

$$v_f = \frac{v_o X_{C2}}{X_{C1} + X_{C2}}$$

or,

$$\beta = \frac{v_f}{v_o} = \frac{1/j\omega C_2}{1/j\omega C_1 + 1/j\omega C_2}$$

$\therefore$

$$\beta = \frac{C_1}{C_1 + C_2} \quad (16-54)$$

The gain of a CB amplifier is given by,

$$A_v = \frac{h_{fe} R_C}{h_{ib}} = \frac{R_C}{r_e}$$

According to Barkhausen criteria

$$A_v \beta \geq 1 \text{ or, } \frac{R_C}{r_e} \frac{C_1}{C_1 + C_2} \geq 1$$

or,

$$\boxed{\frac{R_C}{r_e} \geq \frac{C_1 + C_2}{C_1}} \quad (16-55)$$

Example 16-11

Design a crystal oscillator like Fig.16-33 to generate a signal with 8 MHz frequency. Consider the value of stray capacitance is 3.5 pF, $V_{CC} = 12$ V, and $h_{fe} = 200$.

Solution:

From datasheet it is found that the required load capacitance of an 8 MHz crystal is 20 pF. Let $C_1 = C_2 = 33$ pF. So the load capacitance will be $C_T = C_1 \parallel C_2 + C_s = 33 \text{ pF} \parallel 33 \text{ pF} + 3.5 \text{ pF} = 20 \text{ pF}$ (as required).

From Equ.(16-54),

$$\beta = \frac{33 \text{ pF}}{33 \text{ pF} + 33 \text{ pF}} = \frac{1}{2}$$

So the gain of the CB amplifier should be greater than 2. We can design the CB amplifier exactly in the same method of Example 15-5. Only the

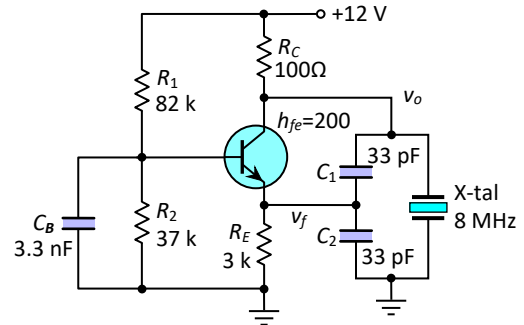


Fig.16-34: Crystal oscillator using CB amplifier

difference is, here we need $A_v \geq 2$. In Example 15-5, the value of r_e was 26Ω .

$$\therefore A_v = \frac{R_C}{r_e} \text{ or, } R_C = 2 \times 26 \Omega = 52 \Omega$$

To make the voltage gain a little greater we can set $R_C = 100 \Omega$. The value of C_B can be calculate but a good approximation is $C_B = 100 \times C_1 = 3.3 \text{ nF}$.

Comments:

16.19 Pierce crystal oscillator

Pierce crystal oscillator is a modification of Colpitts oscillator. Here the inductor of Colpitts oscillator has been replaced by a crystal. The pierce crystal oscillator is shown in Fig.16-35(a). The feedback process is illustrated in Fig.16-35(b).

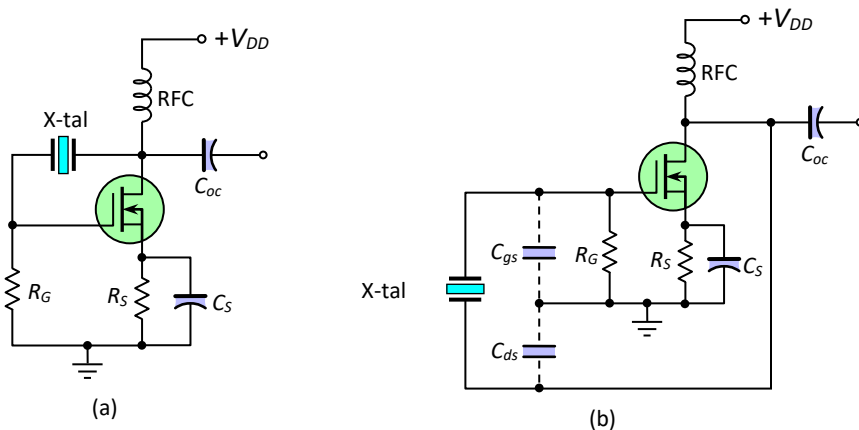


Fig.16-35: Pierce crystal oscillator (a) Normal circuit, and (b) Circuit with FET junction capacitances

The gate-source capacitance (C_{gs}) and the drain-source capacitance (C_{ds}) come across the crystal. Here the crystal works in series resonance mode. In series resonance the crystal is equivalent to an inductor. This inductance of the crystal, and the parasitic capacitances of the FET form a tank circuit. Rest of the operation is same as the Colpitts oscillator.

16.20 Pierce Crystal Oscillator Using Inverter

Pierce crystal oscillator circuit is most commonly used to generate CLK (clock) signals in digital circuits. In that case a CMOS inverter is used as an inverting amplifier. It is the most common CLK generator because of the ease of implementation inside an IC, it has good stability, high Q (ranging from 10,000 to several hundred thousand), and low manufacturing cost. As shown in Fig.16-36, a Pierce crystal oscillator primarily comprises a logic gate inverter which acts as an amplifier, a feedback resistor (R_f) between the inverter's input and output, and an external crystal component. This arrangement is known as a parallel resonant circuit.

The inverter block is a simple CMOS logic inverter used as a basic building block of digital integrated circuits. In conventional mode, its transfer characteristic is non-linear as shown in Fig.16-37. Usually, a CMOS inverter is so designed that its output changes when the input changes near $V_{CC}/2$, that is, the inverter gives the maximum gain (inverted) when the input is near $V_{CC}/2$. The inverter by itself provides a phase shift of 180° . To bias the input of the inverter the feedback resistor is used. With this feedback resistor, both the NMOS and PMOS of the inverter will be on with equal channel resistance. So, the output will be just $V_{CC}/2$. Due to very large input impedance of MOS transistor, current flowing through R_f will be almost zero. So the input and output voltage of the inverter will be same. At the gates, $V_{CC}/2$ works like the operating point of a normal amplifier. Now if the input voltage increases by a very small amount, PMOS will turn OFF and NMOS will turn ON. So the output will be LOW. Similarly, if the input voltage decreases by a very small amount from $V_{CC}/2$, PMOS will turn ON and NMOS will turn OFF. So the output will be HIGH. Thus the CMOS will work as an amplifier.

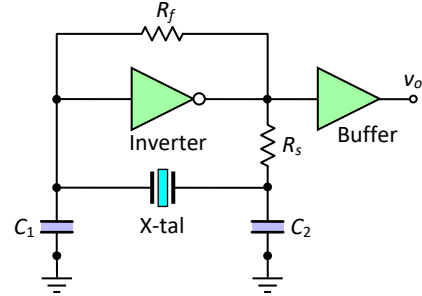


Fig.16-36: Pierce crystal oscillator using CMOS inverter

The inverter used in oscillator circuitry is designed so that the inverter bias point and its maximum gain falls near to half of the supply voltage. Thus, the feedback resistor overcomes the non-linearity of the inverter and transforms a simple logic gate into an analog amplifier. For lower frequencies, the value of R_f is higher in comparison to the higher frequency. For frequency in between 20MHz to 40MHz, typically typical value of R_f is 450 k Ω to 1 M.

Resistor R_s is called the drive limiting resistor, as it is used to protect the crystal from over driven (excess power than rated) by the output of the inverter. However, a large value of R_s will delay the startup of oscillation. R_s also isolates the frequency determining network (crystal, C_1 and C_2) from the output impedance of the amplifier. A thumb rule to select the value of this resistor may be $R_s = X_{C2}$.

A crystal will oscillate at its rated frequency, if the rated load capacitance is connected with it. The values of load capacitance for different crystal are given in their datasheets. The expression to calculate the load capacitance is,

$$C_{load} = \frac{(C_1 + C_{in})(C_2 + C_{out})}{(C_1 + C_{in}) + (C_2 + C_{out})} + C_{st} \quad (16-56)$$

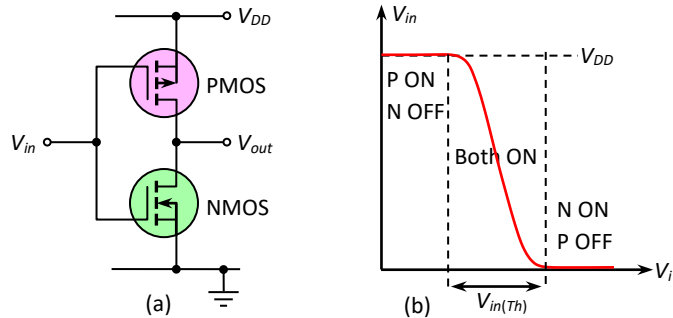


Fig.16-37: (a) CMOS inverter and (b) I-V characteristics

Here, C_{in} and C_{out} are the input and output capacitance of the amplifier and C_{st} is the PCB stray capacitance. Generally the values of C_{in} and C_{out} are given in the datasheet and their value ranges from 2 pF to 5 pF. We can consider $C_{st} \approx 2$ pF.

The amplifier produces 180° phase shift and another 180° phase shift is generated by the frequency selecting network. At first the output signal will lag across C_2 by an amount less than 90° (say 75° degrees). Now the crystal will work as an inductor and in combination with C_1 the rest of 180° , (in this case 105°) phase shift will be produced. Thus, the total phase shift around the loop will be 360° . As this type of oscillator is used to generate square wave, maximum gain of the amplifier is used irrespective of the feedback factor.

This type of oscillator is commonly used in microcontroller ICs.

All the components, except the crystal and C_1 and C_2 , are internally made inside the microcontroller (see Fig.16-38). To set a particular CLK frequency, we have to connect a crystal of rated frequency with the appropriate value of C_1 and C_2 .

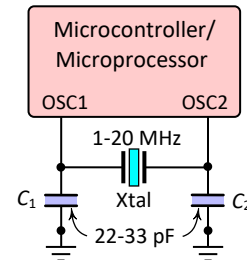


Fig.16-38: Pierce crystal oscillator in microprocessor

Example 16-12 Some digital clocks/watches use a Pierce crystal oscillator of Fig.16-. The required frequency of oscillation is 32786 Hz. This frequency is divided by 15 T-FFs and a 1 Hz ($32786 \text{ Hz} \div 2^{15}$) CLK signal is generated. Design a Pierce crystal oscillator to generate a signal of 32786 Hz frequency.

Solution:

To design the oscillator, we will use 3X8X crystal that is made to generate 32786 Hz signal. From its data sheet (appendix-), we find that the required load capacitance (C_{load}) is 12.5 pF.

Let us consider $C_{in} = C_{out} = 3$ pF, $C_{st} = 0$, and $C_1 = C_2$. Now using Equ.(16-56),

$$12.5 \text{ pF} = \frac{(C_1 + 3)(C_1 + 3)}{(C_1 + 3) + (C_1 + 3)} + 0$$

$$\text{or,} \quad C_1 = 22 \text{ pF} = C_2$$

As the frequency is comparatively low, we can consider $R_f = 1 \text{ M}$ to 10 M . Let it be 2 M

The value of R_s can be calculated as,

$$R_s = X_{C2} = \frac{1}{2\pi f C_2} = \frac{1}{2\pi \times 32768 \times 22 \times 10^{-12}} = 220 \text{ k}$$

Answer: The designed circuit is shown in Fig.16-39.

Comments: We can also use the components values as given in datasheets (Appendix-).

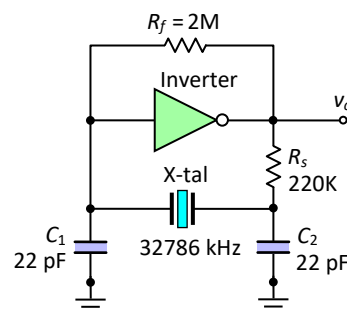


Fig.16-39: Pierce crystal oscillator using CMOS inverter

16.21 Negative Resistance Oscillator

In a tank circuit due to the loss in the resistance of the inductor a damped oscillation is developed. In feedback oscillator, the amplifier continuously compensates this loss and produces an undamped oscillation. Instead of using amplifier, a negative resistance device can also be used to compensate the loss of the inductor. Just opposite to a positive (normal) resistance, a negative resistance can deliver energy. Tunnel diode, SCR, IMPATT diode etc., show the negative resistance in their I-V characteristic curves. Depending on the connection of the negative resistance device with the tank circuit, there are two types of oscillators- **negative resistance series resonant oscillator** and **negative resistance parallel resonant oscillator**. The conceptual diagrams of these oscillators are shown in Fig.16-40. In both circuits, the negative resistance device has to be biased at the middle of its dynamic negative resistance region as shown in Fig.16-41(a).

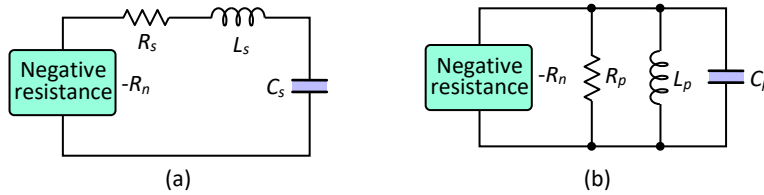


Fig.16-40: Negative resistance oscillator: (a) Series, and (b) Parallel

To compensate the loss of the tank circuit, the value of the negative resistance should be greater than the damping or load resistance in the tank circuit. From the Fig.16-40(a), the total impedance of the series-resonant oscillator is,

$$Z_s = -R_n + R_s + j(\omega L_s - 1/\omega C_s)$$

If $R_n = R_s$, the resistive part, that produces loss, will be zero. So, a continuous oscillation will be produced. The frequency of oscillation can be found by equating the imaginary part to zero.

$$j(\omega_o L_s - 1/\omega_o C_s)$$

$$\omega_o^2 = \frac{1}{L_s C_s}$$

$$\omega_o = \frac{1}{\sqrt{L_s C_s}} \quad (16-57)$$

Similarly from Fig.16-40(b), the total impedance of the parallel-resonant oscillator is,

$$Z_p = -R_n \parallel X_{Cp} \parallel R_p \parallel X_{Lp}$$

$$\frac{1}{Z_p} = \frac{1}{-R_n} + \frac{1}{X_{Cp}} + \frac{1}{R_p} + \frac{1}{X_{Lp}}$$

$$\frac{1}{Z_p} = \frac{1}{-R_n} + j\omega C_p + \frac{1}{R_p} + \frac{1}{j\omega L_p}$$

Here also, if $R_n = R_p$, the resistive part will cancel out and an undamped oscillation will be produced. The frequency of oscillation can be found by equating the imaginary part to zero.

$$\begin{aligned}
 j\omega_o C_p + \frac{1}{j\omega_o L_p} &= 0 \\
 \omega_o^2 &= \frac{1}{L_p C_p} \\
 \omega_o &= \frac{1}{\sqrt{L_p C_p}}
 \end{aligned} \tag{16-58}$$

16.22 Tunnel Diode Oscillator

Tunnel diode was invented in August 1957 by Leo Esaki, Yuriko Kurose, and Takashi Suzuki when they were working at Tokyo Tsushin Kogyo, now known as Sony. In 1973, Esaki received the Nobel Prize in Physics, jointly with Brian Josephson, for discovering the electron tunneling effect used in these **diodes**¹. The symbol and I-V characteristics of a tunnel diode are shown in Fig.16-41. When voltage across the diode increases, current through the diode increases due to the *tunneling effect*. This tunneling effect occurs up to V_P . Values of V_P ranges from 0.1 V to 0.3 V. Beyond this voltage the tunneling effect decreases and so does the tunneling current. As with the increase of voltage current decreases, this portion of the characteristics (*A* to *B*) is called the negative resistance. Negative resistance region is found in the voltage range of 0.1 V to 0.7 V. After the valley point, *B*, diode current increases again with voltage as normal PN junction diode.

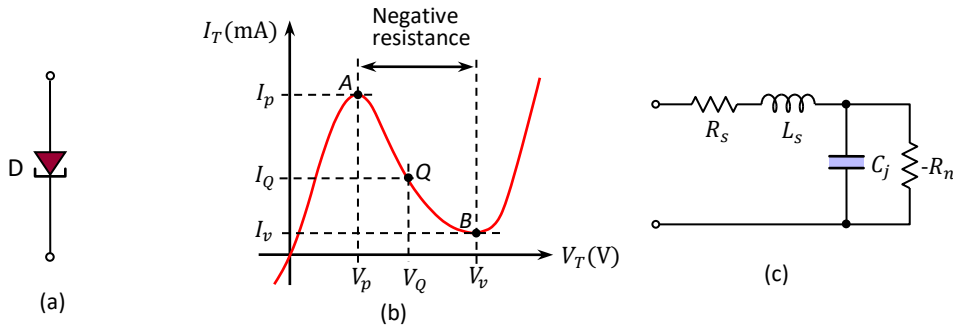


Fig.16-41: Tunnel diode: (a) Symbol, (b) I-V characteristics, and (c) Equivalent circuit

The equivalent circuit of the tunnel diode is shown in Fig.16-41(c). Here L_s is the lead inductance effective at very high frequency, R_s is the bulk resistance of the device. C_j represents the junction capacitance and $-R_n$ is the negative resistance offered by the diode.

Tunnel diode is used to design oscillator circuit with very high frequency up to 10 GHz. To understand the oscillation mechanism of current in a tunnel diode, let us consider the circuit of Fig.16-42(a). Using resistor R_L , the operating point is set at the middle of the negative resistance region. As the switch is ON, current I_T through the diode starts to rise so does the diode voltage V_T . At about $V_T = V_P$, current is maximum (point A). After point A, current decreases and the resistance of the tunnel diode increases. Beyond point B, current increases and the voltage across the diode decreases, as $V_T = V - I_T R_L$. So the operating point moves from B to A. Once the Q-

point reaches point A , current decreases again and hence V_T increases. So the operating point moves from point A to point B . In this way current in the circuit (I_T) oscillates from I_p to I_v and the voltage across the diode (also across R_L) oscillates from V_p to V_v .

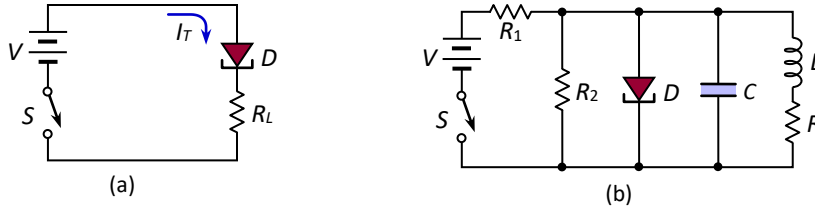


Fig.16-42: Negative resistance oscillator: (a) without tank circuit, and (b) with tank circuit

To control the frequency of oscillation of this voltage (or current) a tank circuit is used in parallel with the tunnel diode as shown in Fig.16-42(b). Here, resistors R_1 and R_2 sets operating voltage and current of the diode. R is the internal resistance of the inductor. The oscillation frequency will be set to the resonant frequency of the tank circuit. The inductor has some resistance that causes the loss of energy. But the negative resistance of the tunnel diode cancels this positive resistance and minimizes the loss.

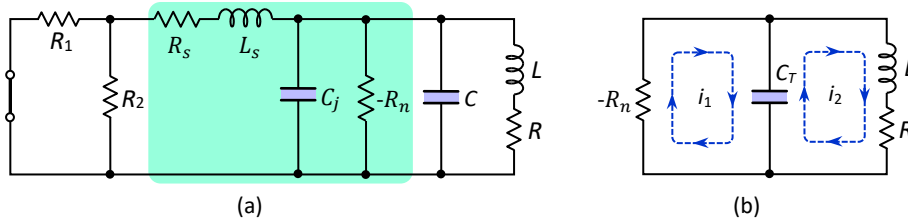


Fig.16-43: (a) Equivalent circuit of negative resistance oscillator, and (b) Simplified equivalent circuit

To analyze this oscillator, let us draw its AC equivalent circuit as in Fig.16-43(a). As L_S and R_S are very small we can neglect them. Generally $R_1 \parallel R_2$ very large, so we can also neglect this resistance. Now assuming $C_T = C_j + C$, we can redraw the circuit of Fig.16-43(a) as in Fig.16-43(b). From this figure, the equations of two mesh-current will be,

$$(-R_n + 1/j\omega C_T)i_1 - (1/j\omega C_T)i_2 = 0$$

and,

$$(1/j\omega C_T)i_1 - (1/j\omega C_T + j\omega L + R)i_2 = 0$$

We can eliminate i_1 and i_2 by equating the determinant of these equations to zero. The determinant of these equations will be,

$$\begin{vmatrix} (-R_n + 1/j\omega C_T) & -(1/j\omega C_T) \\ (1/j\omega C_T) & -(1/j\omega C_T + j\omega L + R) \end{vmatrix} = 0$$

or,

$$-(-R_n + 1/j\omega C_T)(1/j\omega C_T + j\omega L + R) + (1/j\omega C_T)^2 = 0$$

$$\text{or, } (-R_n + 1/j\omega C_T)1/j\omega C_T + (-R_n + 1/j\omega C_T)j\omega L + (-R_n + 1/j\omega C_T)R - (1/j\omega C_T)^2 = 0$$

$$\text{or, } -R_n/j\omega C_T + (1/j\omega C_T)^2 - j\omega R_n L + L/C_T - R_n R + R/j\omega C_T - (1/j\omega C_T)^2 = 0$$

$$\text{or, } -R_n/j\omega C_T - j\omega R_n L + L/C_T - R_n R + R/j\omega C_T = 0$$

$$\therefore j(R_n/\omega C_T - \omega R_n L - R/\omega C_T) + L/C_T - R_n R = 0$$

Equating the real part to zero, we get,

$$\frac{L}{C_T} - R_n R = 0$$

$$\text{or, } R_n = \frac{L}{RC_T} \quad (16-59)$$

Equating the imaginary part to zero and putting $\omega = \omega_o$, we get,

$$j\left(\frac{R_n}{\omega_o C_T} - \omega_o R_n L - \frac{R}{\omega_o C_T}\right) = 0$$

$$\text{or, } \omega_o^2 R_n L = \frac{R_n}{C_T} - \frac{R}{C_T}$$

$$\text{or, } \omega_o^2 = \frac{1}{LC_T} - \frac{R}{R_n LC_T}$$

Putting the value of R_n from Equ.(16-59) into the above equation,

$$\omega_o^2 = \frac{1}{LC_T} - \frac{R^2}{L^2}$$

$$\text{or } \omega_o^2 = \frac{1}{LC_T} - \frac{1}{(L/R)^2}$$

The Q -factor of the coil should be very large. That is $Q = 2\pi L/R$ is very large. So, L/R will also be large. Hence, the second term of the above equation can be neglected. Therefore,

$$\omega_o^2 = \frac{1}{LC_T}$$

$$\text{or, } \omega_o = \frac{1}{\sqrt{LC_T}}$$

$$\therefore \boxed{f_o = \frac{1}{2\pi\sqrt{LC_T}}} \quad (16-60)$$

16.23 Amplitude Stabilization of Oscillators

To design an oscillator we must satisfy Barkhausen criteria, that is $A_v\beta \geq 1$. For any practical oscillator $A_v\beta$ must be greater than 1. Otherwise, the output will not build up. However if $A_v\beta > 1$, the output waveform becomes distorted. In BJT oscillator some sort of self adjustment is achieved due to its nonlinearity, but oscillator using operational amplifier produces most distortion

because of constant gain. To minimize distortion and to keep output amplitude to a desired level, *amplitude stabilization* circuit must be added with these oscillators. The amplitude stabilization circuit controls the gain of the amplifier. At the time of output build up when the amplitude is small, the amplifier's gain remains high so that $A_v\beta > 1$, and when the output achieves a suitable desired amplitude the amplifier's gain is adjusted so that $A_v\beta = 1$. Some common amplitude stabilization circuits are discussed below.

16.23.1 Diode Stabilization Circuit

Fig.16-44 shows a phase shift oscillator with diode amplitude stabilization circuit. Up to ± 1.2 V, the diodes will work as an open circuit. So the resistor R_4 will not be effective in this circuit. Therefore, the voltage gain of the amplifier will be $-R_f/R_3$. When the amplitude of the signal becomes greater than ± 1.2 V, the diodes will be forward biased, and R_4 will come in parallel with the feedback resistor R_f . For this case, the gain of the amplifier will be $-(R_f \parallel R_4)/R_3$ which is less than $-R_f/R_3$. In this way, the gain of the amplifier can be controlled by changing the value of R_4 . Here, two diodes are used in series to make the forward bias voltage 1.4 V, and two such branches have been used in parallel, so that R_4 works for both positive and negative half cycles of the generated signals.

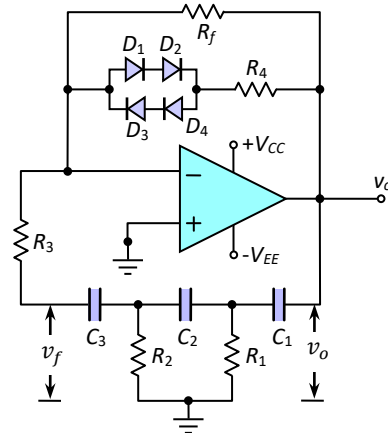


Fig.16-44: Diode amplitude stabilization in RC phase shift oscillator

16.23.2 JFET Stabilization Circuit

Fig.16-45 shows a Wien bridge oscillator with JFET amplitude stabilization circuit. In this circuit the channel resistance (r_{ds}) of the JFET will be added with the input resistor (R_3) of the operational amplifier. So the gain of the amplifier will be $1 + R_f/(r_{ds} + R_3)$. Diode D_1 works as a rectifier and passes only the negative half cycles of the output signal. Capacitor C_3 is a filter capacitor for this rectifier. The negative voltage of this rectifier circuit is applied to the gate of an N-channel JFET. From the characteristics of JFET, we find that it has different g_m at different values of V_{GS} . $1/g_m$ is the drain-to-source resistance of the JFET. For higher values of $|V_{GS}|$, g_m will be low and hence the drain-to-source resistance will be high. Now look at the circuit, when the output amplitude is high, higher voltage will be applied to the gate of the JFET by the rectifier. So the value of r_{ds} will be high and the gain of the operational will be reduced. For low amplitude output signal, the

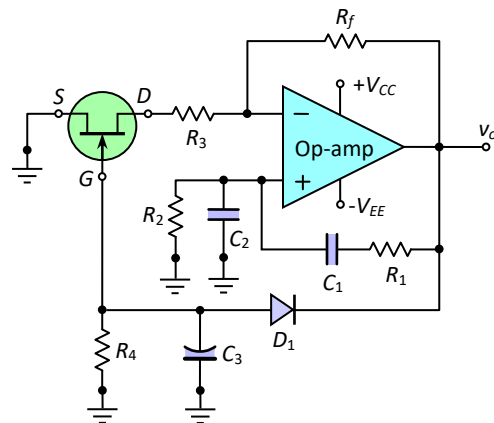


Fig.16-45: JFET amplitude stabilization in Wien bridge oscillator

rectifiers output will be low, hence the gate voltage will be low which results in lower r_{ds} . Therefore, the gain of the operational amplifier will be high. In this way, this oscillator circuit will have $A_v\beta \geq 1$ during the buildup of oscillation, and when the signal achieves a substantial amplitude, $A_v\beta$ will be adjusted to 1. Resistor R_4 is used to discharge the capacitor. Its value should be very large.

References:

1:

https://en.wikipedia.org/wiki/Barkhausen_stability_criterion#:~:text=In%20electronics%2C%20the%20Barkhausen%20stability,linear%20electronic%20circuit%20will%20oscillate.&text=It%20is%20widely%20used%20in,to%20prevent%20them%20from%20oscillating.

1:

[Wikipedia] https://en.wikipedia.org/wiki/Tunnel_diode

2. https://en.wikipedia.org/wiki/Clapp_oscillator

Appendix-A





ECS tuning fork type crystals are used as a clock source in communication equipment, measuring instruments, microprocessors and other time management applications. Their low power consumption makes these crystals ideal for portable equipment.

ECS-3X8X, 2X6X, 1X5X

32.768 KHz Tuning Fork Crystal

[Request a Sample](#)



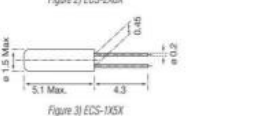
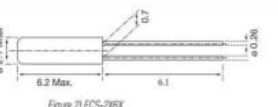
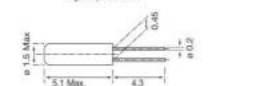
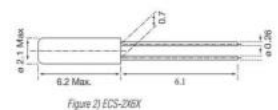
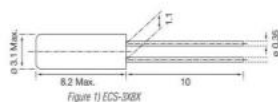
OPERATING CONDITIONS / ELECTRICAL CHARACTERISTICS

ECS-3X8X, 2X6X, 1X5X

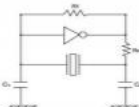


- Cost Effective
- Tight Tolerance
- Long Term Stability
- Excellent Resistance and Environmental Characteristics
- Pb Free/RoHS Compliant

DIMENSIONS (mm)



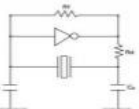
RECOMMENDED OSCILLATION CIRCUIT



ELECTRICAL CHARACTERISTICS

IC: TC 4069P

RECOMMENDED OSCILLATION CIRCUIT



ELECTRICAL CHARACTERISTICS

IC: TC 4069P

Rf: 10MΩ

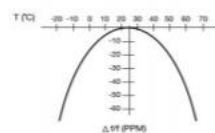
Rd: 330KΩ (As required)

C₁ = 22pF, C₂ = 22pF

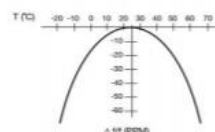
V_{DD} = 3.0V

In this circuit, low drive level with a maximum of 1μW is recommended. If excessive drive is applied, irregular oscillation or quartz element fractures may occur.

PARABOLIC TEMPERATURE CURVE



PARABOLIC TEMPERATURE CURVE



To determine frequency stability, use parabolic curve. For example: What is the stability at 45°C?

- 1) Change in T (°C) = 45 - 25 = 20°C
- 2) Change in frequency = -0.04 PPM × (ΔT)² = -0.04 PPM × (20)² = -16.0 PPM

PART NUMBERING GUIDE:

Manufacturer	Frequency	Load Capacitance	Package Type*
ECS	.327	12.5	8X
ECS	.327	12.5	13X
ECS	.327	12.5	14X

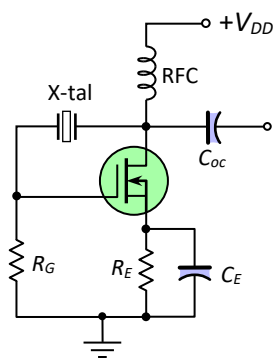
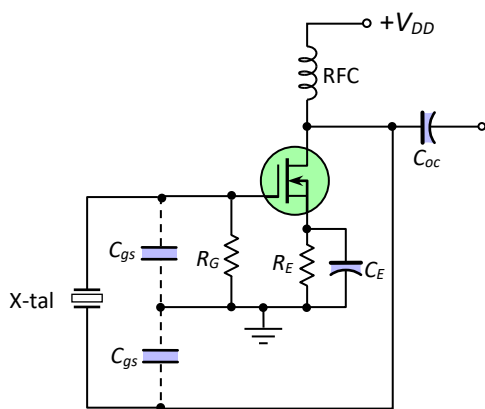
* Package type examples (8X = 3x8, 13X = 2x6, 14X = 1x5)

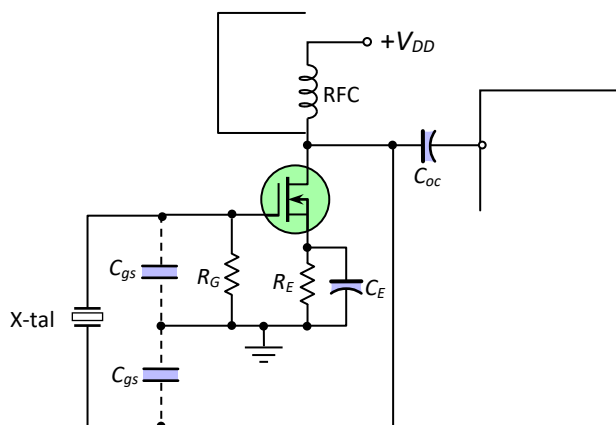
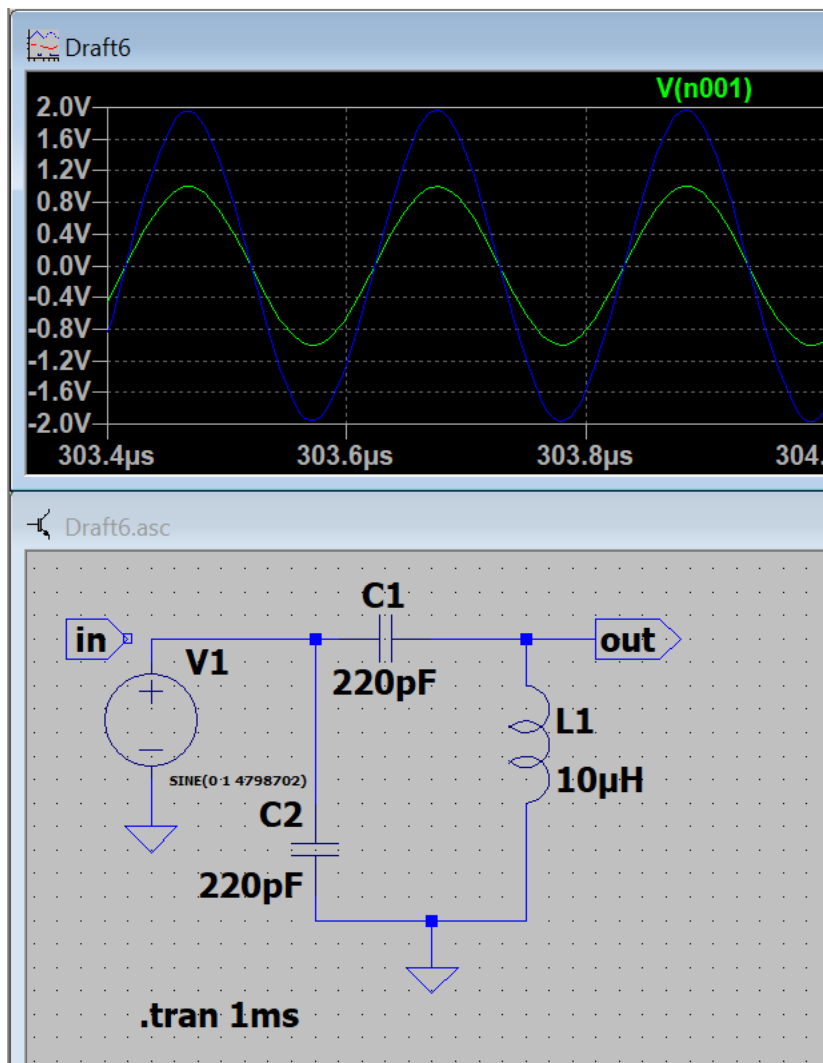
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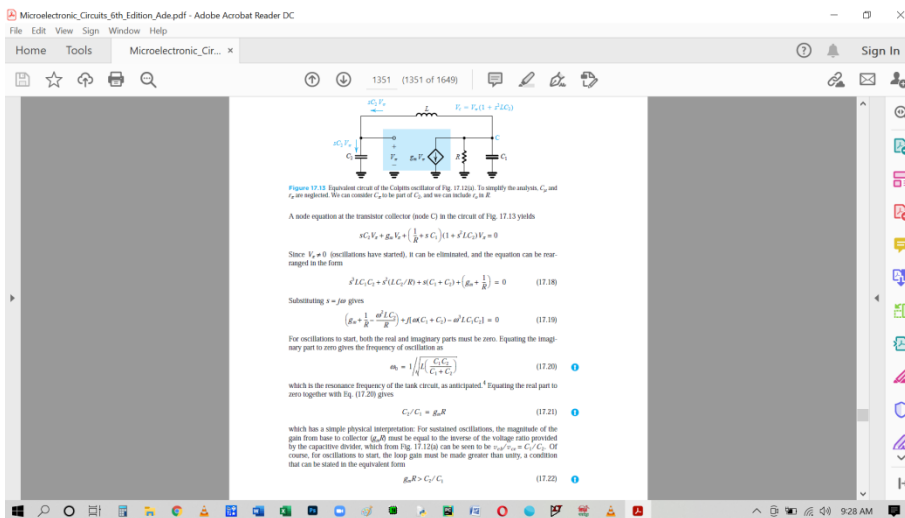
○ +
। না না ধা পা । ক্ষা পা গা ক্ষা । পা ঠ পা ক্ষা । গা রা সা সা ।
○ +
। সা নি ধা নি । সা রে সা নি । গা ক্ষা পা ক্ষা । গা রে সা সা ।

অন্তরাঃ

○ +
। পা ধা নি ধা । সা ঠ সা ঠ । নি রে গা রে । নি রে সা সা ।
○ +
। গা রে সা নি । ধা পা ক্ষা পা । গা ক্ষা পা ক্ষা । গা রে সা সা ।







CAPACITOR STANDARD VALUES (10%)

pF	pF	uF	uF	uF	uF	uF
10	100	0.01	0.1	1	10	100
12	120					
12	120					
15	150	0.015	0.15	1.5	15	150
18	180					
22	220	0.022	0.22	2.2	22	220
27	270					
33	330	0.033	0.33	3.3	33	330
39	390					
43	430					
47	470	0.047	0.47	4.7	47	470
56	560					
68	680	0.068	0.68	6.8	68	680
82	820					

Now converting the output current source into voltage source and adding h_{ie} with $(R - h_{ie})$ we can redraw the circuit for mesh analysis as in Fig.14-3.

